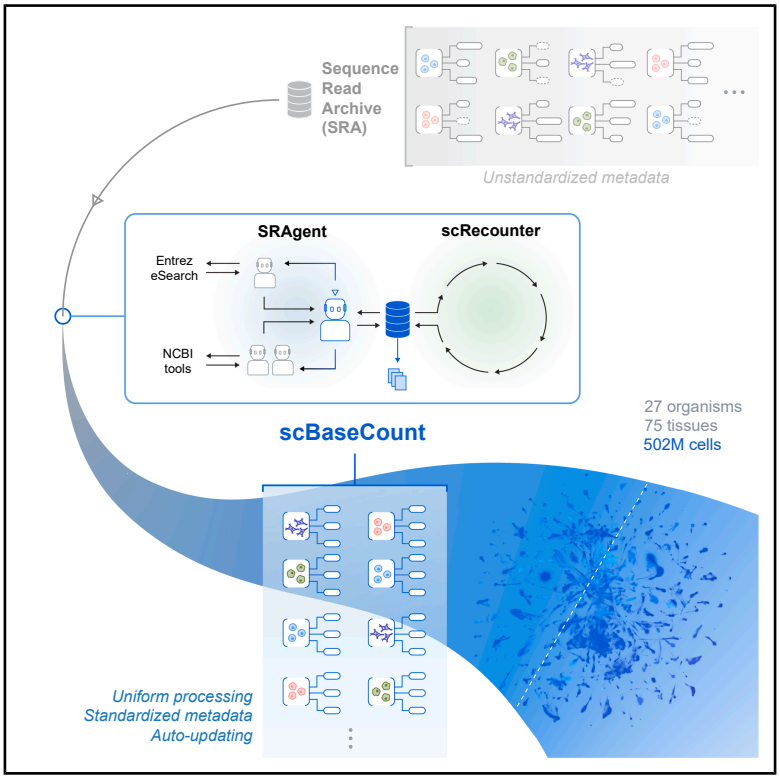
Cell

Resource

scBaseCount: An AI agent-curated, standardized, auto-updated single-cell data repository

Graphical abstract



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In brief

scBaseCount is presently the largest public single-cell RNA-seq repository, containing over 502 million cells across 27 organisms and 75 tissues. An AI agent autonomously discovers, annotates, and uniformly reprocesses all 10× Genomics datasets in the SRA, creating a harmonized, continually updated resource for studying the diversity of cell biology and training AI models.

Highlights

- scBaseCount: the largest public scRNA-seq repository, >502 million cells from SRA
- SRAgent autonomously discovers and annotates all 10× Genomics SRA datasets
- Uniform reprocessing harmonizes exonic, intronic, and non-coding counts
- Spans 27 organisms and 75 tissues for training cell-state AI models

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scBaseCount: An AI agent-curated, standardized, auto-updated single-cell data repository

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SUMMARY

Single-cell RNA sequencing has transformed cell biology by enabling precise transcriptomic measurements of individual cells. The Sequence Read Archive (SRA) is the largest public repository of sequencing reads, yet much of it remains underutilized due to unstandardized metadata. Here, we introduce scBaseCount, a database that leverages an AI agent to automate discovery and metadata extraction and standardize data processing. Built by mining all 10x Genomics datasets, scBaseCount is the largest public repository of single-cell gene expression data, comprising over 502 million cells across 27 organisms and 75 tissues. It offers an unbiased view of the data landscape within the SRA and enables the training of more performant computational models through access to broader phenotypic diversity. Uniform processing enables measurement of both intronic and exonic reads and non-coding gene expression and improves alignment across experiments. Moreover, scBaseCount provides a blueprint for how AI can be leveraged to autonomously curate biological data repositories.

INTRODUCTION

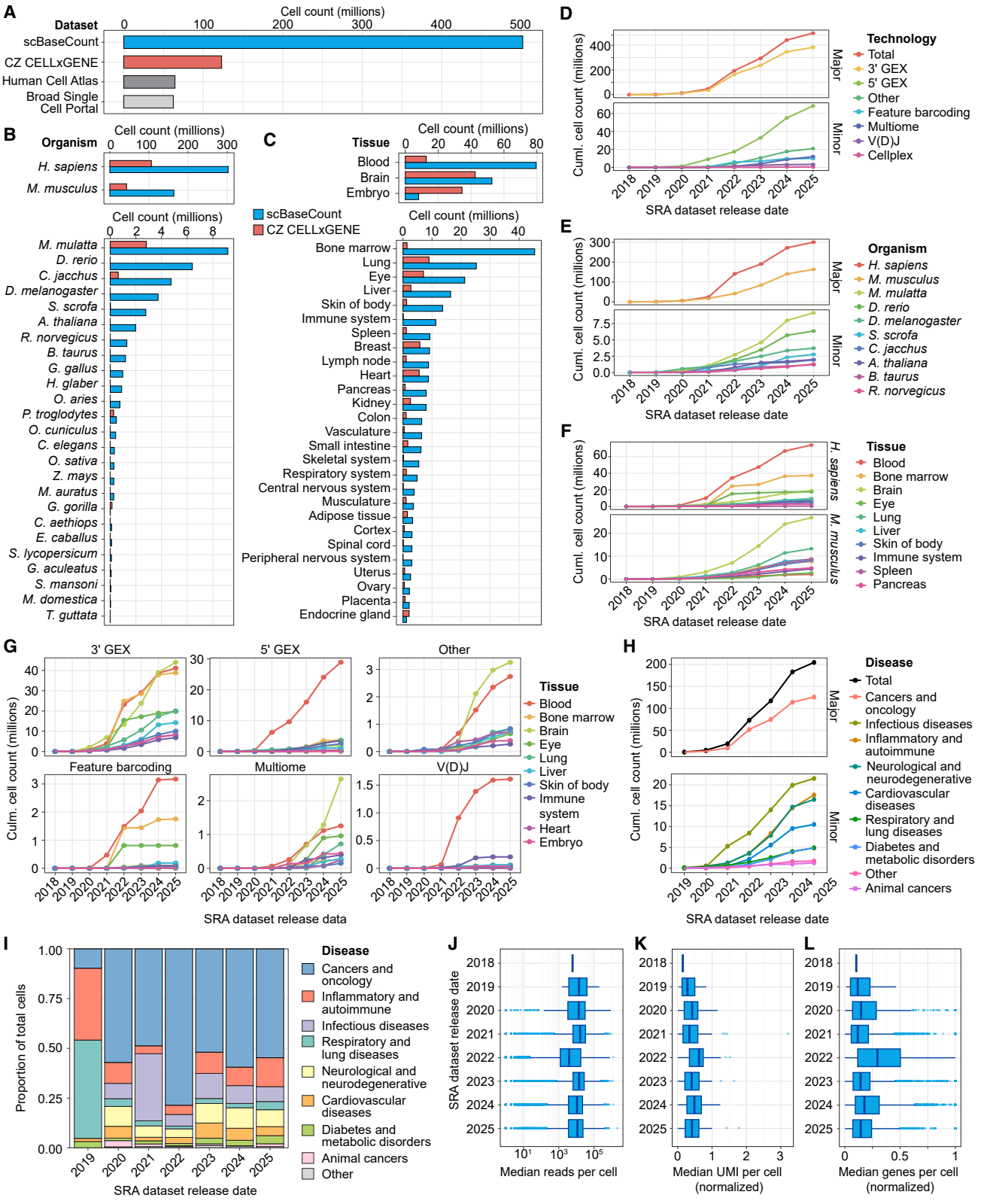
The ability to precisely measure the transcriptomic state of individual cells has transformed the study of cell biology. By uncovering the heterogeneous states of cells as they engage in diverse processes and functions across species and tissue contexts, single-cell RNA sequencing (scRNA-seq) has revealed the finer details of cell identity and behavior that were previously inaccessible through bulk approaches. These discoveries have significantly influenced various fields, from developmental biology to cancer research.[1–3]

Over the past decade, interest in integrating single-cell datasets across institutions and research laboratories has grown significantly. Notable efforts, such as the Human Cell Atlas[4] and the CZ CELLxGENE dataset,[5] have substantially expanded the availability of curated scRNA-seq datasets. These efforts have proven instrumental in advancing our understanding of cell identity, cellular development and differentiation, and disease mechanisms while also providing valuable training data for AI-driven modeling of cellular states. These models are designed to capture context-dependent cellular function and behavior and predict cell state in experimentally unobserved settings, such as in response to perturbation.[6–12] Building these models has emerged as a major frontier in the application of AI to biology.[13,14]

However, initiatives like CELLxGENE and the Human Cell Atlas primarily rely on contributed datasets, which are only a subset of the publicly available data that are accessible through the NIH-hosted Sequence Read Archive (SRA)—the largest repository of raw single-cell sequencing data.[15] While this current approach allows for deeper, expert-level dataset curation and annotation, it also limits the scale and diversity of data available for analysis, particularly for AI models, which typically do not rely on cell labeling. This underscores the need for a new approach to single-cell genomics data curation, one that operates at the scale required for AI model training and is free from the constraints of manual dataset annotation.

Another challenge with existing repositories of single-cell data is that aggregation of datasets from diverse sources introduces analytical variability due to differences in alignment tools,

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reference genomes, and read counting strategies. In the bulk RNA sequencing (RNA-seq) domain, large-scale reanalysis efforts like the ReCount initiative have previously demonstrated the power of standardized pipelines in minimizing these analytical confounders.[16] By uniformly reprocessing bulk RNA-seq data, ReCount provided researchers with a resource where biological variation was not impacted by inconsistencies in data processing pipelines. Drawing on these lessons, we recognized the need for a similar data repository for single-cell genomics, spanning a wide range of species and tissues while adhering to a consistent processing standard. Such a resource should enable more robust cross-study comparisons, facilitate meta-analyses, and better support AI-driven research aimed at modeling cell behavior across diverse biological contexts.

Here, we present scBaseCount, a single-cell genomics data repository that represents the largest-scale reprocessing effort of scRNA-seq data to date. By directly mining all publicly available 10x Genomics datasets from SRA, scBaseCount enables unbiased and comprehensive single-cell transcriptomic analysis of processed data from more than 502 million cells spanning 27 organisms and 75 tissues. Uniform processing enables reduction of processing artifacts and improved analysis downstream, while also producing the largest resource of non-coding transcriptomic data at the single-cell level. When used to train a cell embedding model, scBaseCount enables meaningful improvements in discriminating between diverse phenotypes. scBaseCount is curated by an AI agent and is autonomously updated as more data become available in the repository. This approach opens many new avenues for AI-powered curation and maintenance of large-scale data repositories, offering a generalizable solution to challenges of metadata standardization that currently limit access to many biological datasets.

RESULTS

scBaseCount, the largest public repository of 10x single-cell data, provides novel insights on SRA data composition

scBaseCount is the first comprehensive single-cell database built by directly mining all publicly accessible 10x Genomics scRNA-seq data from the SRA and applying a standardized processing pipeline to improve data harmonization. Using an AI-driven agent, SRAgent, we systematically and continuously identify repositories, unify metadata across diverse sources, and facilitate the discovery, annotation, and reprocessing of raw scRNA-seq data.

To date, SRAgent has processed 208,939 SRA experiments (i.e., SRX entries), of which 105,343 were identified as 10x Genomics sequencing libraries. With the inclusion of 6,059 additional samples currently part of CZ CELLxGENE,[5] we have identified a total of 111,402 target SRX entries at the time of this writing. So far, we have reprocessed 61,381 entries.

Currently, scBaseCount comprises over 502 million cells, with an average of 6,052 unique molecular identifiers (UMIs) per cell. When compared with other large repositories of single-cell data, such as CZ CELLxGENE (123 million cells) and Human Cell Atlas (65 million cells), scBaseCount is the largest collection of publicly available single-cell datasets at the time of this writing (Figure 1A). With data from 27 organisms and 75 tissues, scBaseCount provides a significantly broader range of experimental contexts compared with the current largest repository, CZ CELLxGENE (Figures 1B and 1C).

In addition to dataset identification, SRAgent also attempts to extract key metadata for each SRX, including 10x chemistry, cell versus nuclei suspension type, and associated diseases and tissues. Given the widespread use of SRA as a repository for scRNA-seq data, a comprehensive curation of its contents and associated metadata offers rare insight into broader trends in the field, supported by concrete numbers. We observe that 3′ data have dominated, accounting for 77% of all cells in our collection (Figure 1D). This dominance has remained consistent over the past 7 years, although 5′ data have shown strong growth in the last 2 years (13% of cells in 2024–2025). Among the organisms in scBaseCount, human and mouse cells account for 60% and 32%, respectively (Figure 1E). Notably, mouse cells have increased proportionally 24% from 2023 to 2025, while human cells declined by 29%. Among other organisms, recent trends over the past 3 years indicate increasing representation of *Macaca mulatta* (macaque) and *Danio rerio* (zebrafish). Among tissues, blood and bone marrow are the largest share of the data, while brain data have shown strong growth within the past few years (Figure 1F). Notably, some tissues are strongly contingent on 10x chemistry, such as blood dominating the 5′ and V(D)J data, which makes sense given their relevance to immune profiling (Figure 1G). Among human and mouse datasets, the "Cancers and Oncology" disease category dominated and comprises 30.7% of all cells (Figure 1H). The proportion of cells per disease category has shifted dramatically in some cases between years, such as infectious disease spiking to 33.7% in 2021 from 7.7% in 2020 (a time period overlapping with the peak of the COVID-19 pandemic) and then declining to 5.9% in 2022 (Figure 1I). These data highlight how trends in biomedical research have shifted.

Figure 1. scBaseCount: The largest publicly available repository for single-cell gene expression datasets across species and tissues

(A) Comparison of total cell counts among various cell repositories.

(B) Comparison of cell counts across species in scBaseCount versus CZ CELLxGENE, highlighting a broader species representation and substantially more cells per organism in scBaseCount.

(C) Comparison of cell counts across the top 30 tissues (all mammals) in scBaseCount and CZ CELLxGENE (human and mouse), illustrating the increased tissue diversity and representation in scBaseCount.

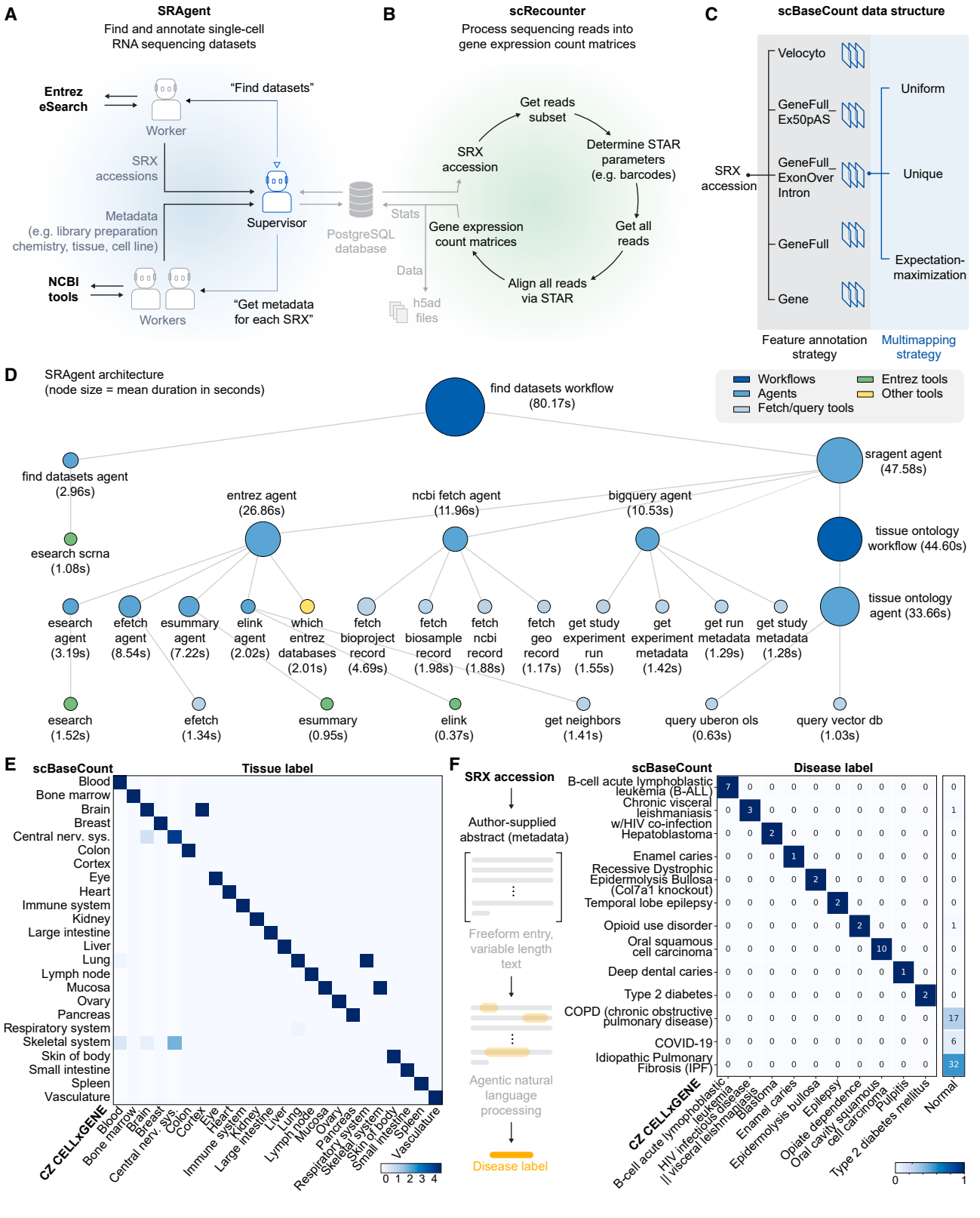
(D–H) Cumulative cell counts by SRA release date grouped by 10x Genomics chemistry (D), organism (E), tissue (F), tissue and 10x Genomics technology (G), or disease (H). Only humans and mice are included in (H). For the sake of clarity, only the top 8 to 10 categories with the most total cells are shown if >10 exist.

(I) The proportion of cells by disease category by year from 2019 to 2025.

(J) The distribution of reads per cell.

(K and L) The distribution of UMIs or genes per cell, normalized to reads per cell.

See also Figure S1 and Table S1.



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While the number of cells has dramatically increased over the past few years, the sequencing depth per cell has been relatively constant (Figure 1J). We found no substantial increase in the number of genes or UMIs per cell after normalizing by sequencing depth (Figures 1K and 1L). The findings suggest that while sequencing costs have declined and sequencer throughput has increased, researchers have opted for sequencing more cells versus increasing the depth per cell.

Automated single-cell data discovery and annotation using SRAgent

To systematically identify and integrate scRNA-seq datasets, we developed SRAgent, a hierarchical agentic workflow built with LangGraph around large language models (LLMs) and specialized tools for querying the SRA. Specifically, SRAgent employs a hierarchical workflow of ReAct agents[17] that asynchronously access eSearch, eSummary, eFetch, eLink, NCBI HTML scraping, SRA BigQuery, sra-stat, and fastq-dump. This workflow continuously mines publicly accessible SRA datasets, retrieves key metadata (e.g., organism, tissue, and disease), and stores these annotations in a relational database (Figure 2A). This automated approach enables rapid discovery of new studies while ensuring metadata curation remains consistent and scalable.

Our simplified diagram of the agent-tool hierarchy illustrates the sophistication of our framework (Figure 2D). While LLMs are rapidly increasing in agentic abilities (METR: https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/), we found that the overall task for SRAgent and the number of tools required were too complicated for a single ReAct agent. We instead decomposed the work into sub-tasks handled by a supervisor-worker agentic architecture. For instance, the find-datasets agent will first call eSearch to obtain new datasets, and then the supervisor calls the SRAgent agent to obtain the target metadata. The SRAgent agent can call the Entrez, ncbi_fetch, bigquery, and tissue_ontology agents. Moreover, the Entrez agent can call a set of sub-agents that each utilize specific Entrez tools. This architecture reduces the task complexity for any given agent and is more token efficient, since only the final result of an agent is passed back to the supervisor.

As noted above, SRAgent has processed 208,939 datasets, of which 105,343 were identified as 10x Genomics sequencing libraries. Assuming a human could process 1 dataset per minute, 3,482 h (145 days) would be required to analyze all 208,939 datasets. We estimate a total token usage of 13.2×10^{12} at a cost of $17,000. By using SRAgent to identify single-cell datasets and then passing them onto scRecounter, we estimate that greater than 0.5 million CPU hours were saved compared with if we had attempted to brute force the task by running every SRX accession through the 10x barcode detector in scRecounter.

When comparing SRAgent's automated tissue annotations to those in CZ CELLxGENE, we found that in most cases, the agent accurately extracted the correct tissue labels (Figure 2E; Table S2). This observation aligns with recent studies demonstrating the effectiveness of LLMs in cell-type annotation.[18–20] While AI models of cell state do not rely on these labels during training, SRAgent's ability to perform reliable tissue labeling enhances confidence in its data curation and annotation capabilities.

In addition to tissue annotations, SRAgent assigns a disease label to each SRX based on the metadata provided by the authors during the SRA submission process (Figure 2F). To validate these disease labels, we used two approaches. First, we compared the labels to those in CZ CELLxGENE for the subset of datasets shared between the two repositories. Because SRAgent labels are derived solely from SRA metadata, they reflect the *condition* under study by the data generator, whereas CZ CELLxGENE typically indicates whether the *individual cells* are diseased. Thus, comparing diseased-cell annotations in CZ CELLxGENE against SRAgent labels provides a reasonable validation, since diseased cells are likely to appear in datasets studying that disease. SRAgent successfully identified the studied disease for all such datasets (Figure 2F; Table S2). We actually found that SRAgent could provide more granularity than in CELLxGENE in certain cases. In particular, we found datasets where all cells labeled as "normal" in CZ CELLxGENE were sometimes associated with a disease according to SRAgent, and manual examination of several such cases revealed strong evidence supporting the assigned disease label (Figures S2–S4). As a second approach, we utilized an LLM-based classifier to generate qualitative confidence scores for the SRX records and found that the majority (51.5%; 107 million cells) were labeled as *high* confidence, while 69.1% (158 million cells) were either *high* or *medium* confidence. Importantly, each LLM classification includes a free-text justification, so researchers can judge whether the scores are valid for their research goals.

In our analysis, we find that using the high-confidence labels yields more accurate annotations at the expense of filtering out

Figure 2. AI-driven curation and standardized processing of scRNA-seq data in scBaseCount

(A) SRAgent workflow: a hierarchical AI-driven pipeline for automated dataset discovery and metadata curation from the SRA. SRAgent systematically queries NCBI tools (e.g., eSearch and eFetch) to identify 10x Genomics datasets, retrieve metadata (e.g., tissue type and library preparation chemistry), and store structured annotations in an SQL database.

(B) scRecounter pipeline: a Nextflow-based workflow for processing raw single-cell sequencing reads into gene expression count matrices. scRecounter downloads and aligns sequencing reads using STARsolo, automatically detects optimal barcode parameters, and generates harmonized expression matrices stored in the h5ad format. Process tracking is managed via a PostgreSQL database.

(C) scRecounter uses multiple feature annotations and multi-mapping strategies to generate a variety of CELLxGENE count tables. Users can choose the option that best suits their application. We currently provide access to all variations in scBaseCount.

(D) Simplified diagram of the hierarchical agentic architecture of SRAgent. Node size denotes the average time required per workflow component.

(E) Matching of SRAgent tissue labels with CELLxGENE tissue labels for a shared subset of data.

(F) Matching of SRX disease labels derived from author-supplied abstracts at the dataset level with CELLxGENE disease labels, which are generally assigned to subsets of cells.

See also Figures S2–S5.

many low-confidence, but still correct, annotations (Figures S5A and S5B). In summary, our findings suggest that disease labels can be used confidently for many research applications but reveal that publicly available metadata quality could be improved, possibly via agentic approaches that facilitate metadata submission and validation.

Along with this publication, we have made the SRAgent code publicly available. The research community can access and utilize SRAgent to further expand single-cell dataset discovery and integration efforts. Of note, SRAgent is a general tool that can provide information for many SRA-related queries and includes an assortment of command-line interface (CLI) tools for ease of use.

Standardized data reprocessing using scRecounter

To process the 10x Genomics datasets identified by SRAgent, we developed scRecounter, a custom Nextflow pipeline[21] designed for efficient and scalable single-cell data reprocessing (Figure 2B). To date, 12.4×10^{12} of 13.8×10^{12} total reads were successfully mapped to a reference genome with STARsolo.

In analyzing scRNA-seq data, researchers often make choices regarding gene models and counting strategies, as there is no universal approach that fits all applications. Depending on the specific analysis, whether focused on standard gene expression quantification, alternative splicing, or transcriptional dynamics, different gene models and counting strategies may be more appropriate. With scRecounter, our goal is to provide a comprehensive set of possibilities, enabling researchers to select the most suitable approach for their own study. To achieve this, scRecounter offers multiple feature annotation strategies, including Gene, GeneFull, GeneFull_Ex50pAS, GeneFull_ExonOverIntron, and Velocyto[22] (see STAR Methods). These models vary in their treatment of exonic and intronic regions, allowing users to choose between exon-only counts (Gene), exon and intronic counts (GeneFull), the hybrid between the two that is implemented in CellRanger (GeneFull_Ex50pAS), or a combination of the two for RNA velocity analysis (Velocyto) (Figure 2C).

Additionally, scRecounter is flexible in terms of multi-mapping strategies, which determine how multi-mapped reads are handled. These include expectation maximization (EM),[23] which probabilistically assigns multi-mapped reads, as well as unique and uniform strategies, which either retain only uniquely mapped reads or distribute multi-mapped reads evenly (Figure 2C; see also Figures S6–S8).

By incorporating these diverse options, scRecounter ensures that scBaseCount remains a versatile resource, accommodating a broad range of scRNA-seq applications while minimizing technical biases. Along with this publication, we have also made the scRecounter code available.

Clustering and annotation of scBaseCount data

Unlike existing single-cell data repositories and atlases, cell annotations in scBaseCount were not manually assigned by independent experts. This presents unique challenges in interpreting the data, understanding its diversity, and exploring its finer structure beyond what is provided in the original SRA annotations. A deeper analysis of the dataset requires comprehensive annotation of cell categories across its entirety. To address this, we developed a computational approach to annotate all data contained within the repository.

Given the diversity of species in the scBaseCount data, we required a cell-type annotation strategy that would scale well to our number of cells and could robustly see through the technical noise from different modalities, different source labs, and different disease contexts. For this reason, we embedded the cells using State,[12] a state-of-the-art cell embedding model that was pre-trained on scBaseCount and has been shown to outperform existing models in differentiating between subtle phenotypes. In order to annotate cell types, embeddings generated for human cells using State were annotated using a cell-type classifier trained on data from Tabula Sapiens.[24] Our classifier strategy assigned each cell in scBaseCount to the best matching cell type in Tabula Sapiens within the same tissue, using the tissue labels provided by SRAgent. By utilizing both the tissue information and the transcriptional profile, this strategy helps to eliminate artifactual cell-type categorizations.

When visualizing the distribution of classified cells across a random sampling of SRXs from scBaseCount, we observed clustering of shared cell types, such as leukocytes and muscle cells, across diverse tissues (Figure 3A). Across the full dataset, leukocytes, particularly T and B cells, were overrepresented in nearly all tissues (Figure 3B). This enrichment may reflect sampling biases across studies or a preponderance of diseased samples enriched for immune cell populations. Other commonly observed cell types include epithelial and endothelial cells.

To validate our cell-type assignments, we used two complementary approaches. First, we evaluated classifier performance using a confusion matrix comparing predicted and true cell labels in held-out CZI collections (Figure S6). Second, we assessed whether our annotations could recapitulate known marker genes.

For the marker gene recapitulation analysis, we applied a differential expression framework based on t-tests to random tissue-specific subsets of scBaseCount (see STAR Methods). Using this approach, we recovered a high proportion of independently curated marker genes from PanglaoDB.[25] Importantly, marker recovery in scBaseCount was competitive with that of Tabula Sapiens, even though our classifier did not explicitly use marker genes and the curated marker set was independent of Tabula Sapiens annotations (Figure 3C).

Overall, the overlap between known cell-type markers and those recovered from scBaseCount supports the accuracy of both our classifier and the dataset. In particular, the shared marker gene signatures with PanglaoDB indicate that scBaseCount is of sufficient experimental quality to recover biologically meaningful genes and provides a representation of human cell biology comparable to Tabula Sapiens. We have made these marker gene annotations available as a GMT file for the field to use (Data S1).

Given our ability to successfully annotate human cells, we then looked to similarly annotate cell types from other species that are well represented in scBaseCount and have high-quality reference atlases. We found that *Drosophila*, zebrafish, and mice all had readily available, whole-organism reference atlases as well as many cells in our repository. With this in mind, we used

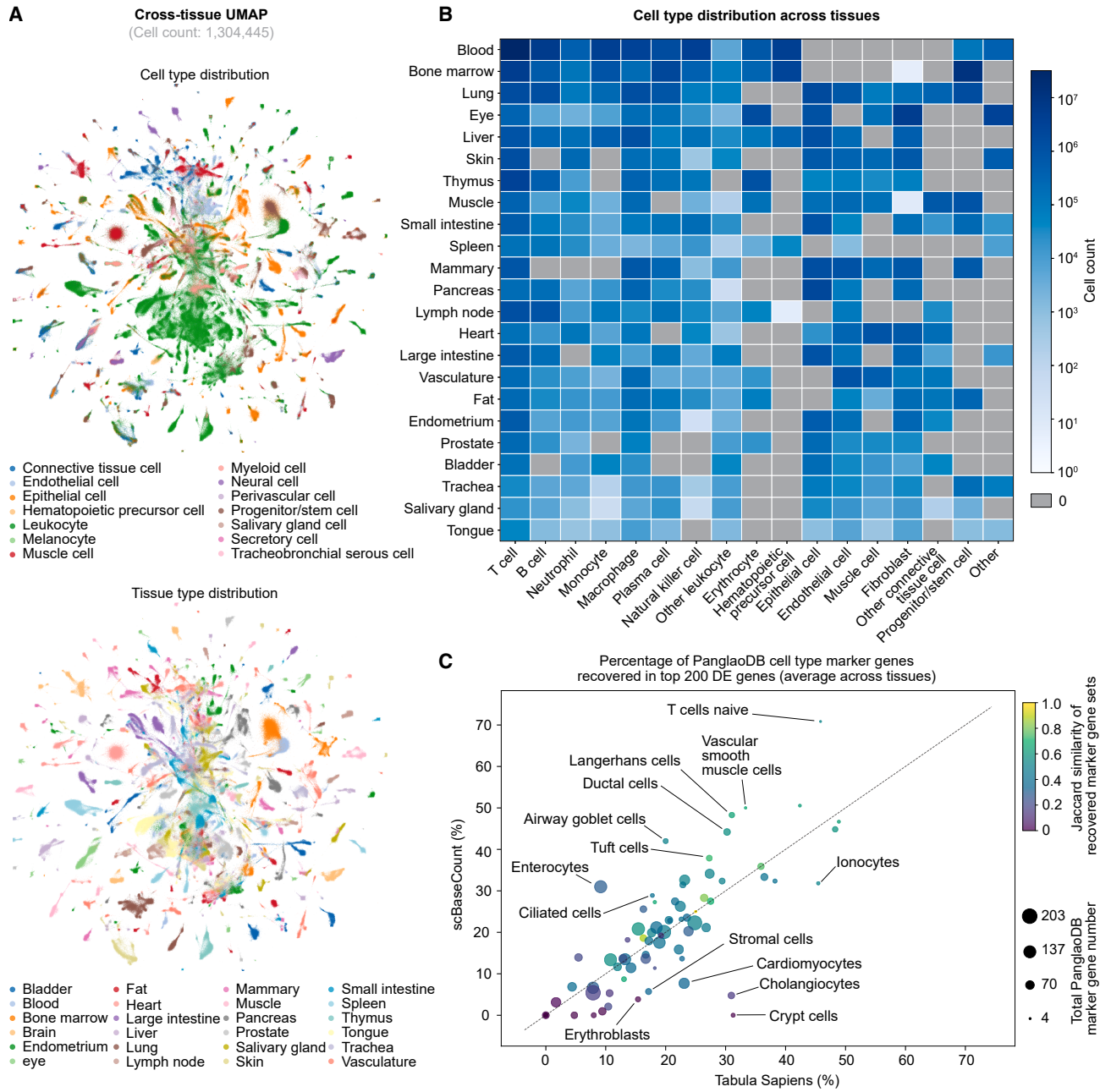


Figure 3. Cell-type annotation and clustering of data in scBaseCount

(A) Uniform manifold approximation and projection (UMAP) visualization of random per-tissue samples of approximately 50,000 annotated human cells each in scBaseCount (N = 1,304,445) using state embeddings. Outlier cells beyond the 0.1 percentile–99.9 percentile range for both x and y axes coordinates are excluded from the plot. Top panel: cells colored by predicted cell type; bottom panel: cells colored by tissue label.

(B) Distribution of inferred cell types by tissue category.

(C) Scatterplot comparing the percentage of PanglaoDB marker genes recovered among the top 200 upregulated genes for each cell-type category between scBaseCount and Tabula Sapiens, averaged across tissues. DEGs were computed on random samples of approximately 500,000 annotated human cells for every tissue.

See also Figures S9–S11.

Scanorama[26] to label cells from each of these species in a tissue-specific manner (see STAR Methods). Interestingly, the cell-type distributions of these species reflect that different model organisms are adopted by different fields, with *Drosophila* data skewing toward neurons and zebrafish being overrepresented in precursor stem cells (Figure S7). As a demonstration

of one potential use case for these markers, we converted genes to human homologs across species, computed the Jaccard overlap between marker gene sets for all pairwise combinations of all cell types in all 4 species, and used these overlaps to derive a cross-species cell-type adjacency matrix that highlights conserved and divergent cell-type relationships (Figure S8). This brief, macro cell-type clustering is a demonstration of the many biological use cases of scBaseCount. As a resource for the field, we have made the marker genes computed over these cell types available as GMT files (Data S2–S4).

Uniform data reprocessing reduces the contribution of analytical confounders

A key advantage of a uniform processing pipeline for single-cell transcriptomic data is its ability to minimize technical variability introduced by differences in analytical pipelines. To assess the amount of signal in our data originating from technical factors, we used silhouette scoring,[27] a metric that quantifies how well individual data points cluster based on a given categorical variable. The silhouette score ranges from -1 to 1, where higher values indicate that cells are more cohesively grouped within the same category and well-separated from other categories, while lower values suggest overlapping or poorly defined clusters. We applied this approach to evaluate how various categorical metadata variables influence gene expression patterns in scBaseCount (Figure 4A). Some factors primarily reflect biological variation (e.g., tissue type), while others capture a mix of technical and biological variation (e.g., sample suspension type, library preparation chemistry, or sample ID). Our goal was to quantify the extent to which groupings based on each factor shape the dataset's structure and contribute to observed variation in gene expression. In scBaseCount, technical factors such as library preparation chemistry and sample suspension type (single-cell versus single-nucleus sequencing) exhibited comparable or lower silhouette scores than biologically meaningful categories like tissue type when compared with CZ CELLxGENE, which is composed of pre-processed datasets that are not uniformly processed (Figure 4B). For example, in the case of sample ID (largely equivalent to the SRA study or SRP ID in scBaseCount) (Figure S9), the silhouette scores are similar to tissue type in scBaseCount, while in CZ CELLxGENE, they are significantly higher, suggesting that study-specific processing introduces stronger batch effects (Figure 4B).

This effect is even more pronounced for sample suspension type, which differentiates single-cell from single-nucleus sequencing. We reasoned that this may likely be due to the common practice of counting single-nucleus data using both exonic and intronic reads (i.e., pre-mRNA annotations), while single-cell data typically includes only exonic reads. While neither approach is inherently incorrect, this inconsistency could introduce strong batch effects when integrating both data types. scBaseCount mitigates this issue by retaining and reporting both exonic and intronic read counts, resulting in better integration of single-nucleus and single-cell datasets.

To further examine this possibility, we focused on four shared datasets, listed in Table S3, that contain both single-cell and single-nucleus sequencing data. We performed principal-component analysis (PCA) and visualized the datasets in two-dimensional PC space (Figure S10A). In the 2-dimensional PC space, tissue type showed a higher silhouette score in scBaseCount than in CZ CELLxGENE, and conversely, sample suspension type (i.e., cells versus nuclei) scored lower in scBaseCount (Figure S10B). Examining the source publications of these four integrated datasets, we found that the single-nucleus data in this subset was indeed processed with pre-mRNA references, and the single cells were processed with a standard exonic reference, as is common practice (Table S3). Correcting this technical artifact in scBaseCount resulted in better separation of tissue types along the first two PCs compared with CZ CELLxGENE (Figure S10C).

Beyond harmonization, uniform data processing in scBaseCount, including consistent treatment of exonic and intronic reads across samples, enables additional downstream analyses, such as RNA velocity, to better model transcriptional dynamics and cell state transitions. As mentioned in the previous section, scRecounter provides cell×gene count tables for spliced and unspliced transcripts as well using Velocyto.[22]

Impact of scBaseCount on training AI models of cell state

Next, we evaluated the impact of scBaseCount when used as training data for AI models of cell state, specifically for the task of generating an embedding of the cell that captures meaningful biological variation across diverse contexts. Specifically, we trained a 100M-parameter state embedding model[12] on a filtered subset of scBaseCount (N = 214 million cells) and CZ CELLxGENE (N = 134 million cells). State embedding is an encoder-decoder model that uses a transformer to construct a dense representation of a cell from its gene expression values and an MLP to reconstruct expression from the dense embedding. After training each model on human cells from scBaseCount and CZ CELLxGENE, we obtained two models, State_bsc and State_cxg, which we then used to embed cells from several held-out datasets. We trained classifiers to distinguish between phenotypes (see STAR Methods), using only the cell embeddings from State_bsc and State_cxg, at varying levels of granularity (Figure 4C).

At the coarsest level, we evaluated embeddings on cell-type classification, as this is the most distinguishable phenotype. For this task, we used the Tabula Sapiens dataset,[24] which includes data from 24 donors spanning 180 cell types. Both State_bsc and State_cxg achieved very high performance (approximately 90%), though the model trained on scBaseCount showed a modest improvement.

We then tested the models on subtler state changes, such as those induced by disease or perturbation. For disease, we used datasets containing cells infected by COVID-19.[28–30] For perturbation, we used CRISPRi gene knockdown experiments for 2024 essential genes[31,32] across four cell lines (HepG2, RPE1, Jurkat, and K562). In both cases, State_bsc clearly outperformed State_cxg, improving classification accuracy by 2.24% for infection status and 23.9% for perturbation group.

Finally, to probe even finer phenotypic resolution at the level of individual genes, we compared differential gene expression predictions. Performance on this metric is a strong indicator of a model's ability to capture perturbation effects, a core

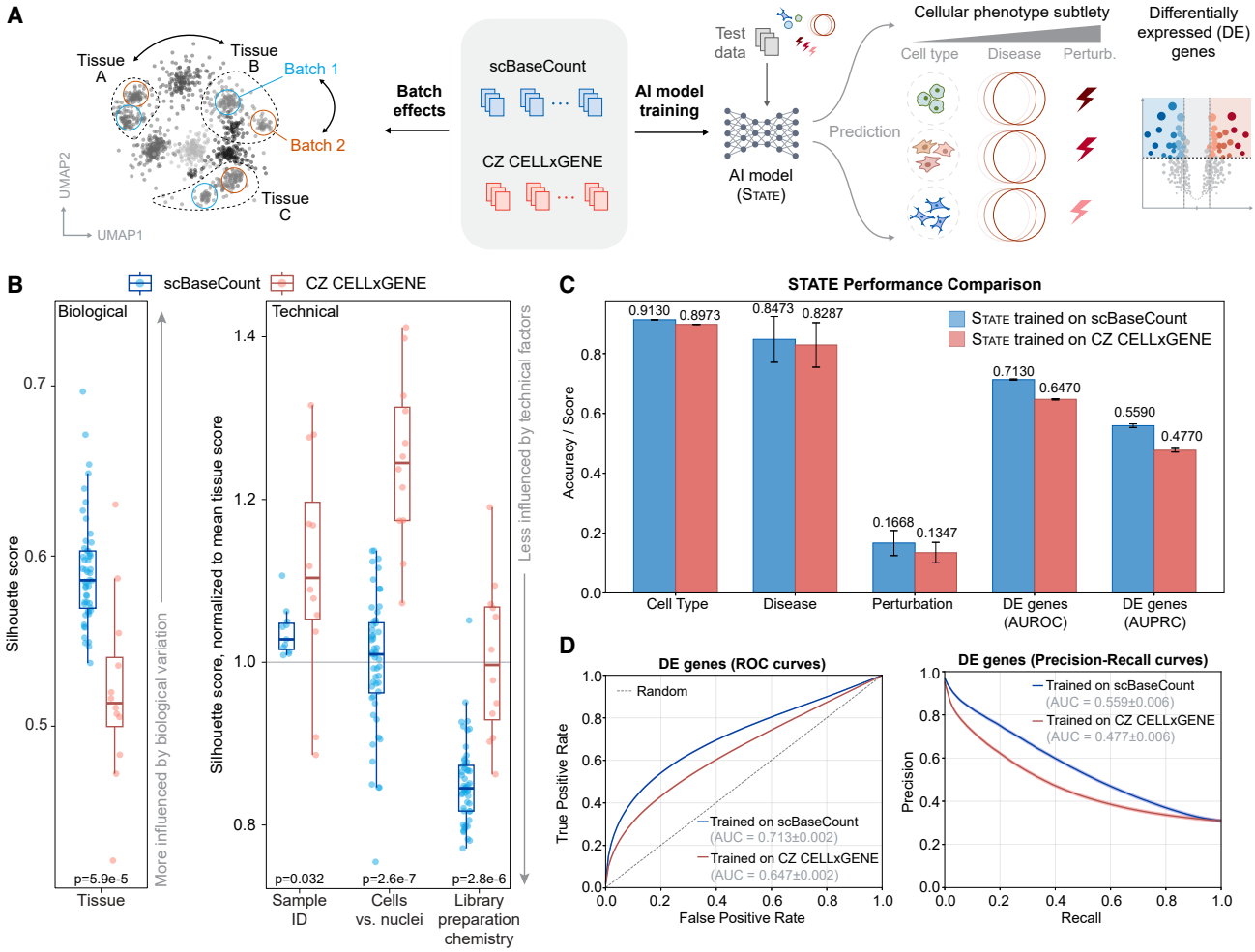


Figure 4. Impact of scBaseCount on downstream analyses and AI model training

(A) Comparing the impact of technical and biological factors on clustering in scBaseCount and CZ CELLxGENE and on downstream AI model training.

(B) Silhouette scores computed across random subsets of 250,000 cells from scBaseCount (blue) and CZ CELLxGENE (red) for different metadata factors. Higher scores indicate a greater influence of the corresponding factor on data clustering. For technical factors, scores are normalized to the mean tissue score for each dataset. CZ CELLxGENE subsets were drawn from 20 individual human datasets and downsampled to 250,000 cells in two steps, and scBaseCount subsets were randomly sampled from human and mouse accessions. The resulting objects were filtered to retain genes present in at least 10 cells and cells with at least 30 detected genes. All datasets were sum-normalized, log-transformed, and processed using PCA before calculating silhouette scores (scib_metrics package). These results demonstrate that, given the contribution of tissue type, technical factors such as sample ID, library preparation chemistry, and sample suspension type contribute less to clustering in scBaseCount compared with CZ CELLxGENE, highlighting how scBaseCount's uniform processing pipeline reduces analytical variability while preserving biological structure. *p* values were computed using a two-sided Mann-Whitney U test.

(C) Performance difference in probing cell embeddings from the State model trained on scBaseCount ($N = 214$ million cells) and State trained on CZ CELLxGENE ($N = 134$ million cells). Embeddings are probed with a multilayer perceptron to classify specific conditions, like cell type, disease, or perturbation group. Error bars represent the standard deviation; the value above each bar is the mean.

(D) Performance difference in reconstructing DEGs from the State model trained on scBaseCount versus that trained on CZ CELLxGENE. True DEGs, determined as those with FDR < 0.05 using a Wilcoxon rank-sum test, are compared with DEG predictions from State model reconstructions (see STAR Methods). Curves are generated by varying the threshold for calling significance from model outputs.

requirement for virtual models of cell state and behavior. Here, the benefit of training on scBaseCount was most striking. We conducted a Wilcoxon rank-sum test across all perturbation conditions to get a ranked set of differentially expressed genes (DEGs) for each perturbation. We obtained ground truth DEGs by filtering for false discovery rate (FDR) < 0.05. Next, data from the genetic perturbation screens were embedded, and the State Decoder was used to reconstruct the expression for all cells. We obtained ranked lists of DEGs from the State model reconstructions by ordering genes according to the difference in the reconstructed expression for perturbed groups compared with control groups. Receiver operating characteristic (ROC) and precision-recall (PR) curves were obtained by varying the sensitivity threshold to call DE (see STAR Methods). State_bsc showed increased sensitivity to subtle gene expression changes (10.2% improvement in area under the ROC curve [AUROC] and

17.2% improvement in area under the PR curve [AUPRC] over State_cxg) (Figure 4D).

To further assess the impact of uniform processing on model training, we retrained State_bsc and State_cxg while restricting the training set to the exact same source FASTQ files. Under this controlled setting, both models performed comparably, with no meaningful difference in performance (Figure S11). This indicates that State_bsc's performance improvement was driven mainly by the increased number of cells rather than uniform processing. However, a uniformly processed, large-scale single-cell repository can uniquely enable the development of new AI model architectures that leverage cellular context when generating embeddings. Because uniform processing of SRA data produces a separate expression count matrix for each SRX (SRA experiment), data remain aligned with donor, experimental batch, sequencing lane, and other covariates. This structure enables training tabular foundation models such as Stack[33] that leverage within-experiment cellular context to improve downstream performance. Such experiment-specific modeling would not be feasible at the same scale using existing single-cell data repositories, where data are not consistently separated by experimental sample, and this information must be inferred from user-provided metadata, which is often incomplete. Indeed, when contextual information is removed, Stack shows a significant drop in performance.[33]

DISCUSSION

In this study, we present scBaseCount, a continuously updated and uniformly processed multi-species scRNA-seq data repository that is built through directly mining all publicly accessible 10x Genomics datasets. Our motivation stems from the need to create a large, high-quality resource for training computational models of cell state and behavior. Existing single-cell databases rely on manually curated datasets, which limits their size and diversity, failing to leverage the vast reserves of publicly available transcriptomic data. Moreover, integrating individual datasets within these repositories remains a major challenge due to variation in how each of these datasets was processed.

By systematically mining, annotating, and processing raw single-cell transcriptomic reads directly from SRA, scBaseCount is designed to be a large, harmonized repository of single-cell datasets with minimal analytical confounders. Inspired by the ReCount initiative, which demonstrated the value of uniformly processed bulk RNA-seq data,[16] scBaseCount reduces technical artifacts across studies, ensuring that AI models learn biologically meaningful variation. It also far exceeds other single-cell databases in species diversity, reflecting the breadth of phenotypic information it captures. Since it contains non-coding transcriptional data for all cells, it is currently the largest repository of such data at single-cell resolution.

We believe that these benefits are particularly relevant in the context of training AI models of cell state, where dataset size, species diversity, and disease diversity are crucial to model performance. In our experiments, even when restricting to only human data, training models with scBaseCount results in a marked improvement in a model's ability to distinguish subtle but important differences in transcriptional state, such as perturbation and infection. Notably, this does not capture the additional performance gains likely to result from scBaseCount's unprecedented species diversity, as demonstrated in prior work.[10,11] Beyond serving as a drop-in replacement for existing datasets, scBaseCount also enables new model architectures. For example, its inclusion of intronic counts across all cells allows modeling of RNA transcriptional dynamics, such as RNA velocity,[22,34] at an unprecedented scale.

scBaseCount is the first large biological data repository curated by an AI agent. Our automated workflow, powered by the SRAgent AI system and a Nextflow-based processing pipeline, ensures consistent metadata curation and analytical treatment across all datasets. This agentic workflow has the advantage of being entirely automated, easily scalable, and capable of continuously updating as new data become available. Furthermore, this workflow is able to utilize the abstracts associated with data uploads to create meaningful annotations, which are helpful for both biological interpretation and data processing.

In the time since it has been released, scBaseCount has already been used to generate testable biological hypotheses. So far, many of these insights have come from its use in pretraining foundation models,[33,35–37] including the use of RNA-velocity-based approaches.[38] We believe that scBaseCount also has uses for assembling atlases of rare diseases, cell types, and species.[39] As demonstrations of this, we have published example notebooks showing how scBaseCount can be used to easily create an atlas of Huntington's disease, as well as how scBaseCount can be used to query embryonic data across many species (see data and code availability). We hope that by reducing the workload in identifying and pre-processing single-cell datasets from under-studied contexts, scBaseCount will help facilitate research in these areas.

Limitations of the study

Despite these advantages, several challenges remain. While the agentic workflow automates much of metadata extraction, some tasks such as cell-type annotation may require human expertise, community-sourced curation, or additional tools for mapping individual cells onto annotated references.[6,40–42] Our automated cell type annotations are further constrained by the cell types present in the reference atlases used for classification. Moreover, high-quality whole-organism references for non-human species are available for only a subset of organisms in scBaseCount. Other cell-level annotations such as the applied perturbation or donor information are also not accessible through SRA-derived metadata and must be incorporated manually. Similarly, because SRAgent extracts disease annotations from study-level abstracts rather than structured per-sample metadata, a single disease label may be propagated to all samples within a study, including healthy controls.

In the current release, we have focused on libraries created using the 10x Genomics platform and sequenced on established Illumina sequencers. While our focus on 10x Genomics ensures technical consistency across datasets, these library preparation methods exhibit known systematic cell-type biases.[43–45] As alternative library preparation methods become more commonplace and sequencing technologies continue to evolve, we must adapt scBaseCount to maintain compatibility with new

modalities, including various chemistries, multi-omic measurements, and spatial transcriptomics. Although our curation to date has prioritized publicly accessible 10x datasets, expanding to other platforms and protected datasets is a key goal. This requires manual requests for access, but we are actively engaging with authors and anticipate that remaining gaps with other repositories will gradually close. Looking ahead, we also plan to extend scRecounter to support additional single-cell technologies, further broadening scBaseCount's scope.

Conclusion

Ultimately, this unified and continuously updated resource provides a foundation for AI-driven efforts to model cell behavior across health and disease. By delivering the largest uniformly processed single-cell data repository, scBaseCount lowers barriers for large-scale model training and integrative analysis. While our focus here is on single-cell applications, many other fields could benefit from centralized, uniformly processed repositories, particularly where standardized metadata are lacking. We believe that AI-driven data curation and automated processing present a compelling new framework for unlocking such datasets. Thus, scBaseCount represents not only a step forward for single-cell modeling but also a blueprint for building large, uniformly processed repositories in other areas of biology.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Hani Goodarzi (hani.goodarzi@arcinstitute.org).

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

- All single-cell data generated in this study (the scBaseCount repository) are publicly available as of the date of publication through the Arc Virtual Cell Atlas https://github.com/ArcInstitute/arc-virtual-cell-atlas/. This paper also analyzes existing, publicly available datasets, whose accession numbers and resource identifiers are listed in the key resources table.
- All original code (SRAgent, scRecounter, and the analysis code) is publicly available as of the date of publication.
  - SRAgent: https://doi.org/10.5281/zenodo.21539103
  - scRecounter: https://doi.org/10.5281/zenodo.21539134
  - scBaseCount_analysis: https://doi.org/10.5281/zenodo.21539161
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Y.H.R., A.D., and H.G. conceived the project. Y.H.R., H.G., A.D., P.D.H., S.K., and D.P.B. supervised the project, with Y.H.R. coordinating effort across the team. N.D.Y. conceived and developed SRAgent. N.D.Y., C.C., R.I., and A.D. developed scRecounter. C.C., N.D.Y., A.N., A.A., Y.H.R., J.P., and N.T. performed data analysis, incorporation of CZ CELLxGENE projects, and comparisons between the two datasets. N.D.Y., C.C., A.N., A.A., and C.R.-T. visualized data. A.A. and R.S. trained AI models for learning representations of cells. Y.H.R., N.D.Y., C.C., A.A., A.N., and H.G. wrote the document, incorporating comments from all authors.

DECLARATION OF INTERESTS

D.P.B. acknowledges outside interest as a Google advisor. H.G. acknowledges outside interest as a co-founder of Exai Bio, Vevo Therapeutics, and Therna Therapeutics; as a member of the board of directors at Exai Bio; and as a scientific advisory board member for Verge Genomics and Deep Forest Biosciences. P.D.H. acknowledges outside interest as a co-founder of Terrain Biosciences, Stylus Medicine, and Spotlight Therapeutics; as a member of the board of directors at Stylus Medicine; as a board observer at EvolutionaryScale and Terrain Biosciences; as a scientific advisory board member at Arbor Biosciences and Veda Bio; and as an advisor to NFDG, Varda Space, and Vial Health. N.D.Y. acknowledges outside interest as a co-founder of Flock Bio. Y.H.R. is a scientific advisory board member at GC Therapeutics.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used Claude (Anthropic) for proposing text, checking for consistency, and converting files to and from LaTeXion of files from LaTeX. After using Claude, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
scBaseCount data repository	This paper	https://github.com/ArcInstitute/arc-virtual-cell-atlas
Tabula Sapiens	Consortium* et al.[24]	https://cellxgene.cziscience.com/collections/e5f58829-1a66-40b5-a624-9046778e74f5
Human Brain Cell Atlas	Siletti et al.[46]	https://cellxgene.cziscience.com/collections/283d65eb-dd53-496d-adb7-7570c7caa443
CRISPRi perturbation screen	Replogle et al.[31]	https://plus.figshare.com/articles/dataset/_Mapping_information-rich_genotype-phenotype_landscapes_with_genome-scale_Perturb-seq_Replogle_et_al_2022_processed_Perturb-seq_datasets/20029387
Extended CRISPRi perturbation data	Nadig et al.[32]	https://plus.figshare.com/articles/dataset/_Mapping_information-rich_genotype-phenotype_landscapes_with_genome-scale_Perturb-seq_Replogle_et_al_2022_processed_Perturb-seq_datasets/20029387
Fly Cell Atlas	Li et al.[47]	https://flycellatlas.org/#data
Tabula Muris	Schaum et al.[48]	https://cellxgene.cziscience.com/collections/0b9d8a04-bb9d-44da-aa27-705bb65b54eb
Zebrafish Cell Atlas	Lange et al.[49]	https://zebrahub.sf.czbiohub.org/cbg?name=cbg_full
CZ CELLxGENE collections	CZI Cell Science Program et al.[5]	https://cellxgene.cziscience.com
10x Genomics reference genomes (human, mouse)	10x Genomics	https://www.10xgenomics.com/support/software/cell-ranger/latest/release-notes/cr-reference-release-notes
ENSEMBL genome assemblies	Frölich et al.[50]	https://www.ensembl.org
Uberon ontology	Mungall et al.[51]	https://www.ebi.ac.uk/ols4/ontologies/uberon
MONDO disease ontology	Vasilevsky et al.[52]	https://obofoundry.org/ontology/mondo.html
OBO Foundry Cell Ontology	Diehl et al.[53]	https://github.com/obophenotype/cell-ontology
PanglaoDB	Franzén et al.[25]	https://panglaodb.se/markers.html?cell_type=%27choose%27
Software and algorithms		
SRAgent	This paper	https://doi.org/10.5281/zenodo.21539103
scRecounter	This paper	https://doi.org/10.5281/zenodo.21539134
scBaseCount analysis code	This paper	https://doi.org/10.5281/zenodo.21539161
STAR / STARsolo	Dobin et al.[54]	https://github.com/alexdobin/STAR
Nextflow	Di Tommaso et al.[21]	https://www.nextflow.io
LangGraph	LangGraph	https://pypi.org/project/langgraph/
LangSmith	LangSmith	https://pypi.org/project/langsmith/
ChromaDB	ChromaDB	https://github.com/chroma-core/chroma
scikit-learn	Pedregosa et al.[55]	https://scikit-learn.org
scanpy	scanpy	https://scanpy.readthedocs.io
scib_metrics	Lücken et al.[56]	https://github.com/YosefLab/scib-metrics/tree/v0.5.8
Scanorama	Hie et al.[26]	https://github.com/brianhie/scanorama
State Embedding model	Adduri et al.[12]	https://huggingface.co/arcinstitute/SE-600M

METHOD DETAILS

SRAgent metadata extraction

SRAgent is a Python package that utilizes LangGraph for constructing the agentic workflows. We used LangSmith for profiling SRAgent runs.

SRAgent is deployed on Google Cloud Platform (GCP) Cloud Run with 2 CPUs and 2 GB of memory per job. To comply with NCBI API rate limits, jobs are triggered every 1-5 minutes, processing 3-5 datasets per run with a peak rate of 300 datasets per hour. The

mean processing time per dataset is 80 seconds, with sub-agents and tools called asynchronously for increased parallelization. All extracted metadata is stored in a GCP-hosted SQL database for downstream processing.

To obtain Uberon tissue ontologies and MONDO disease ontologies for each SRA dataset,[51] we first converted the node text in the Uberon ontology graph to a ChromaDB vector database. SRAgent performs a semantic search with a free text tissue description derived from data obtained via NCBI Entrez tools. For each returned record, the agent then explores neighboring terms in the ontology hierarchy and queries the Ontology Lookup Service (OLS) when additional validation is needed. We note that many SRA experiment datasets could not be disambiguated to just one tissue ontology term based on the metadata in NCBI. In such cases, we retained multiple values.

Disease categories were determined from the MONDO ontology graph.[52] For computational efficiency, the graph was pruned to just ontology nodes found in the scBaseCount dataset, with intermediate pruned nodes replaced with a single edge in order to retain connections among the remaining nodes. We created a weighted adjacency matrix for all remaining nodes, with weights set by the number of corresponding records in scBaseCount. We then applied the scikit-learn[55] spectral clustering algorithm to produce 12 clusters. To create the cluster labels in an unbiased manner, we provided the top 10 ontology IDs per cluster by weight to Claude Opus 4 and prompted for concise cluster labels based on the cluster compositions.

We note that because SRAgent extracts disease annotations from study-level abstracts rather than structured per-sample metadata, a single disease label (e.g., "COVID-19") may be propagated to all samples within a study, including healthy controls. The SRA does not currently provide structured disease metadata at the sample level, and reliably extracting donor-level disease status would generally require mining full-text manuscripts, which is complicated by unreliable linking of metadata to individual donors, PDF parsing challenges, and journal paywalls. To partially address this issue, we implemented an LLM-based approach for determining disease label confidence scores, given the study, experiment, and sample metadata available. First, we obtained key metadata from SRA Entrez and BigQuery, including biosample attributes (e.g., sample label and organism), SRX attributes (e.g., number of biosamples), and study attributes (e.g., associated paper titles and abstracts). We then created a dataset of 150 randomly sampled SRX metadata records, each annotated with a curated, gold-standard disease-label confidence score qualified at three levels: *low*, *medium*, and *high* confidence. We leveraged this dataset to tune the prompt for an LLM classifier (gpt-5-mini) of disease label confidence. The classifier reached high accuracy on the gold standard (recall: 0.90, precision: 0.86). Notably, the classifier provides free-text justification for each confidence score. We then applied the classifier to all SRX records with disease labels generated by SRAgent (N = 40,729).

scRecounter pipeline

scRecounter is a Nextflow[21] pipeline designed for scalable reprocessing of single-cell RNA-seq data. So far, we have focused on processing 10x Genomics datasets as identified by SRAgent.

The pipeline runs on GCP Cloud Run and submits jobs to GCP Batch. To stay within GCP resource quota limits, the pipeline processes a maximum of three datasets per run, with new runs triggered every three minutes. Process tracking is managed via a PostgreSQL database hosted on GCP.

10x chemistry identification

Once SRAgent has predicted that an SRX dataset contains 10x Genomics single-cell data, the dataset is processed through the scRecounter pipeline. Since SRAgent could not consistently determine specific assay chemistry, the first step of scRecounter was to determine the chemistry using a Cell Ranger-inspired automated detection algorithm. The first 1 million reads from each SRX are processed using STARsolo, testing barcode sets corresponding to 10x 5′, 10x 3′ v2, 10x 3′ v3, and 10x Multiome GEX. 10x 3′ v1 was not included. The barcode file yielding the highest percentage of valid cell barcodes was selected. Datasets with fewer than 30% valid barcodes were removed at this stage.

Reference genome preparation

For gene annotations, we used human and mouse reference genomes from the 10x Genomics repository. STAR references for other species were generated using the following workflow: for each species, a widely used ENSEMBL genome assembly was selected and downloaded.[50] As with human and mouse annotations, we applied a custom script to filter each species' GTF file, retaining only relevant biotypes (e.g., "protein-coding" and "lncRNA") based on their clade (e.g., mammals, birds, fungi). This filtering ensures that only genes measurable by single-cell RNA-seq are included and maintains consistency with 10x Genomics annotations. For STAR reference generation, we adjusted the genomeSAindexNbases parameter according to genome size, following recommendations from the STAR developers.[54] These annotations and references are available for download through our portal.

STARsolo data processing

After identifying the correct 10x chemistry, we then aligned all SRRs (SRA runs) from an SRX with a single STARsolo run, using the following parameters:

```
–soloType CB_UMI_Simple
–clipAdapterType CellRanger4
```

```
–outFilterScoreMin 30
–soloCBmatchWLtype 1MM_multi_Nbase_pseudocounts
–soloCellFilter EmptyDrops_CR
–soloUMIfiltering MultiGeneUMI_CR
–soloUMIdedup 1MM_CR
–soloFeatures Gene GeneFull GeneFull_ExonOverIntron GeneFull_Ex50pAS Velocyto
–soloMultiMappers EM Uniform
–outSAMtype None
–soloBarcodeReadLength 0
```

scRecounter leverages `STARsolo's –soloFeatures` parameter, enabling the generation of various count matrices tailored to specific analytical needs:

Gene: Counts reads mapping entirely to exonic regions of annotated transcripts, capturing fully spliced transcripts. This procedure represents traditional gene expression analyses and was the default in earlier `CellRanger` versions.
GeneFull: Counts reads overlapping entire gene locus, i.e. including exonic and intronic regions, which captures both unspliced (primary) and spliced transcripts and also provides a more comprehensive view of gene activity.
GeneFull_ExonOverIntron: Counts reads overlapping exonic and intronic regions, but assigns higher priority to exonic overlaps. This option helps to resolve reads that map to overlapping genes.
GeneFull_Ex50pAS: Similar to the above option, but with more sophisticated priorities scheme, which prioritizes partial and antisense exonic overlap over intronic reads.
Velocyto: Generates separate count matrices for spliced, unspliced, and ambiguous reads, following the rules from.[22] This enables RNA velocity analyses to infer dynamic cellular processes.

For `Gene, GeneFull, GeneFull_ExonOverIntron, and GeneFull_Ex50pAS,` three multi-mapping strategies are used:

Unique: retain only uniquely mapped reads
Uniform: distribute multi-mapped reads evenly
Expectation-Maximization (EM): probabilistically assign multi-mapped reads

`Velocyto generates the following count matrices:`
Spliced: reads mapping exclusively to exonic regions, representing mature mRNA molecules
Unspliced: reads containing intronic sequences, representing nascent pre-mRNA transcripts
Ambiguous: reads that cannot be definitively classified as (un)spliced due to mapping ambiguity

In total, the pipeline generated 15 count matrices per SRX dataset.

CELLxGENE reprocessing

We obtained accession codes corresponding to CZ CELLxGENE collections and re-processed the corresponding SRXs using the method outlined above. Constituent SRXs from each collection were then assembled into one h5ad object per collection, using *Unique GeneFull_Ex50pAS* counts.

We then mapped the observation metadata from each CZ CELLxGENE collection to our re-processed collections by matching cell barcodes. Specifically, within each collection, the cell barcodes follow the format [ATCG]-[10x_well_id], and our STARsolo-processed data follows the same format. Therefore, for each collection, we mapped the 10x_well_ids in the CZI version of the collection to ours by assigning each identifier to the corresponding CZ CELLxGENE identifier that maximized the number of intersecting barcodes. By using this method, we were able to successfully map metadata from CZ CELLxGENE collections into scBaseCount.

Dataset subsampling

For CZ CELLxGENE, each subsampled data contained samples from 20 projects, where all projects were solely 10x assays and labeled as 'primary_data' in CZI metadata. Each dataset was subsampled to 30% of the original before joining. Each sample was subsequently downsampled to 250,000 cells as necessary. For scBaseCount, we selected *GeneFull_Ex50pAs* counts from randomly chosen SRX accessions, and concatenated them together until the object reached 250,000 cells. In both cases, half of the subsets were mouse, and half were human.

State model training

To evaluate the comparative utility of scBaseCount and CELLxGENE for training single-cell foundation models, we trained the State Embedding model[12] using 108 million trainable parameters. The architecture consisted of an eight-layer transformer with an embedding dimension of 1024 for the encoder and a multi-layer perceptron to decode expression. Training was conducted on four GPUs with a batch size of 96 and gradient accumulation of 8, yielding an effective batch size of 3072. All other training details match the primary manuscript.

For the scBaseCount dataset, we applied a filter to retain only human cells, excluding those with fewer than 300 non-zero gene counts or 500 unique molecular identifiers (UMIs), resulting in 214 million cells for model training. The CELLxGENE dataset was not additionally filtered beyond its original curation, resulting in 134 million cells for model training.

Both datasets were standardized to a common gene set comprising 19,790 human protein-coding genes. The State Embedding model utilizes pretrained gene embeddings as input features, an approach that has demonstrated superior performance compared to one-hot encoding schemes.[10,11] We generated gene representations using ESM-2-based featurization, computed by averaging per-amino acid embeddings across all protein-coding transcripts within each gene. To remove dataset size as a confounding factor, models were trained for 100,000 optimization steps rather than for a fixed number of epochs. To ensure unbiased evaluation, Tabula Sapiens data were systematically excluded from both training sets.

Probing cell embeddings

To assess the impact of the training dataset on cell embedding information content, we used State Embedding models, trained on scBaseCount or CELLxGENE, to embed held out datasets. For each dataset, we trained a simple MLP probe to classify different conditions given only the cell embedding, such as the applied perturbation, the cell type, or the disease state. The MLPs contained two linear layers with one ReLU activation. For each task represented in Figure 4C, cells within a condition were randomly split 80% / 10% / 10% into train / val / test splits, and probes were trained for five epochs before reporting final test classification accuracy.

To obtain AUROC and AUPR values, we used the State Embedding model to encode the Replogle-Nadig genetic perturbation screen dataset,[31,32] then used the State decoder[12] to reconstruct gene expression values from cell embeddings. For each perturbation, we computed effect scores as the absolute difference between mean predicted log probabilities (averaged across cells) under perturbed versus control conditions. Ground truth DEGs were identified using the Wilcoxon rank sum test at FDR < 0.05. ROC and PR curves were generated by varying the classification threshold across all unique effect score values in descending order, with genes above each threshold classified as differentially expressed and evaluated against the binary ground truth labels.

Human cell type assignment

We retrieved and reprocessed Tabula Sapiens 10x Genomics data for 24 different tissues: blood, brain, bone marrow, lung, spleen, eye, liver, skin, muscle, small intestine, mammary, pancreas, lymph node, heart, large intestine, vasculature, fat, endometrium, prostate, bladder, trachea, salivary gland, tongue, and thymus. We also separately created a cell atlas for the brain based on a re-processing of the public data from the Human Brain cell atlas.[46] We used the State Embedding 600M (SE-600M) model to embed the Tabula Sapiens and Human Brain atlas data. For each tissue, we trained a separate logistic regression model to classify cell type based on the SE-600M embeddings. The logistic regression model was implemented using sklearn, with the max_iter parameter set to 1000.

To map samples (SRX, ERX) to the appropriate tissue model, we used the corresponding tissue category label from SRAgent. The raw text tissue labels of each sample were originally converted to one of 75 CELLxGENE tissue categories by OpenAI's o3 model, including 'other'. We mapped each of these tissue categories in the metadata to the closest tissue model, and 'other' if it did not fit with any of the tissues. For each sample, we embedded the raw counts with the State Embedding model, and applied the classifier that was trained on the same tissue as the sample, making the assumption that each sample only contains one tissue. For our analysis, we omitted samples that had a mean UMI count per cell of 100 or less, and samples classified as 'other' tissue were not annotated. After these filters, we analyzed 166,381,968 human cells with 143 uniquely predicted cell types.

To visualize the cells across tissues, we randomly retrieved samples amounting to approximately 50,000 cells from each tissue, and we plotted a UMAP of these cells using the State embeddings. We colored one plot by the predicted cell type and one plot by the tissue label. Cells beyond the 0.1–99.9 percentile range for the x- and y-axis coordinates are excluded from the plot to filter out extreme outliers. We used the OBO Foundry Cell Ontology[53] to color the UMAP with broader parent cell type categories that encompass our existing cell type labels.

To biologically validate our annotations, we compared cell type marker genes from PanglaoDB to the differentially expressed genes of our predicted cell types. To do this, we first mapped each of our cell type labels to a relevant cell type category in PanglaoDB. We randomly retrieved samples amounting to approximately 500,000 cells for each tissue in scBaseCount, or all of the cells for that tissue if there were less than 500,000 cells, and ranked the genes of each cell type category using t-test score as implemented in the scanpy rank_genes_groups function. We repeated this process for the cell type categories in each tissue of Tabula Sapiens. For each cell type category, we computed the percentage of PanglaoDB marker genes that were detected in the top 200 upregulated genes. We averaged this percentage for each cell type category across tissues, and we plotted these values from scBaseCount against Tabula Sapiens. The size of each point in the scatter plot reflects the number of markers in PanglaoDB for that cell type category. Each point is also colored by the Jaccard Similarity between the set of marker genes detected in scBaseCount and in Tabula Sapiens for that cell type category.

In addition to our marker gene analysis, we ran our pipeline on a subset of CELLxGENE and compared our cell type annotations to the ground-truth labels. We used the SRAgent metadata to subset the CELLxGENE data to human cells. We also mapped each of the unique tissue category labels to the closest tissue model, and 'other' if there was no clear match or a tissue label did not exist, similar to our process of tissue mapping in scBaseCount. We ran our classifier pipeline on the human cells with available cell type labels and tissue model labels, excluding cells with the 'other' tissue label, resulting in 2,732,573 cells annotated for comparison. Since there

were 170 unique ground-truth cell type labels in our CELLxGENE subset and 77 for our predicted annotations, we used the OBO Foundry Cell Ontology to categorize each of our cell type labels into a broader, parent cell type. We plotted two confusion matrices to compare our annotation pipeline's predicted categories to the true category. The first matrix (2,652,156 cells) covers cell type categories that our classifiers can predict, and the second matrix (80,417 cells) covers cell type categories that were exclusively in our CELLxGENE subset but were not in our Tabula Sapiens training data, and hence the classifier assigned those cells to the best predictable category. The confusion matrices are row normalized.

Non-human cell type assignment

For mouse, zebrafish, and *Drosophila* cell types, we first took the relevant reference atlases,[47–49], then subsetted each atlas into the constituent tissues. For each SRX, we matched the tissue label from SRAgent to one of the defined tissues in the reference atlas using sentence transformers, where we used Euclidean distance between the embedding of the agent tissue and the possible tissue labels to find the closest match. Then, we created an anndata object with both the reference data for that tissue and the query dataset, and used Scanorama[26] to integrate them. Finally, we used a logistic regression model to transfer the cell type labels from the reference to the query. For the fly cell atlas, we used the 'S_annotation_broad' column as the cell type, and for zebrafish, we used 'projection_group' as the tissue. For all other cases, the tissue and cell type were labeled as 'Tissue' and 'Cell Type'.

Agentic versus deterministic metadata extraction

While deterministic tools exist for metadata extraction from the SRA (e.g., ffq and pysradb), we choose to develop an agentic AI method because the deterministic approaches often fail to extract metadata that is embedded in unstructured text.[57,58] For example, the SRA submission abstract must be used in many cases to infer whether the dataset is single cell or bulk RNA-seq because structured metadata fields are often incorrectly labeled: scRNA-seq libraries are commonly mislabeled as "TRANSCRIPTOMIC" instead of "TRANSCRIPTOMIC SINGLE CELL" for the *librarysource* field in the SRA BigQuery metadata table. Below, we provide detailed scenarios in which SRAgent must derive target metadata values from complex, unstructured text. Moreover, we used SRAgent, ffq, and pysradb to obtain all target metadata fields from 100 randomly selected SRX accessions, and we found that ffq and pysradb could only obtain metadata for 2 or 3 of the 12 metadata fields (Table S4).

Dataset: SRX with description ''high-throughput transcriptomic profiling of sorted cell populations''

1. **entrez_agent** → eSearch returns metadata; `library_strategy` = ''RNA-Seq''
2. **sragent_agent** → Queries `SRA BigQuery` records and finds "TRANSCRIPTOMIC" label for the *librarysource* field
3. **sragent_agent** → Queries `eFetch` for detailed record

Why deterministic fails: No structured field indicates single-cell; requires natural language understanding of the methods

1. **[Agent reasoning]**: ''sorted cell populations'' could be bulk or single-cell
2. **ncbi_fetch_agent** → Scrapes NCBI page for additional context
3. **[Agent finds]**: Protocol mentions ''10x Chromium controller''
4. **[Agent concludes]**: is_single_cell = ''yes'', lib_prep = ''10x_Genomics''

Dataset: COVID-19 study where individual samples lack disease annotation

1. **entrez_agent** → Sample attributes show: `tissue`=''lung'', `disease`=''''
2. **sragent_agent** → Cannot determine disease from structured fields
3. **ncbi_fetch_agent** → Retrieves study abstract

Why deterministic fails: Disease is only mentioned in unstructured abstract text, with no mention of disease ontology identifier

1. **[Abstract text]**: ''.samples from patients with severe SARS-CoV-2 infection.''
2. **disease_ontology_agent** → Maps ''SARS-CoV-2 infection'' to MONDO:0100096 (''COVID-19'')
3. **[Agent outputs]**: disease = ''COVID-19''

QUANTIFICATION AND STATISTICAL ANALYSIS

Analytical factor evaluation

In an effort to evaluate the contribution of suspension type to the data structure, we selected four human datasets shared between CZ CELLxGENE and scBaseCount collections that contained both single-cell and single-nucleus data. We identified the matching cells in CZ CELLxGENE using the combination of observation & _joinid and cell_type as unique cell identifiers. We excluded cells with fewer than 100 unique genes and genes observed in fewer than 40 cells. The resulting datasets were then sum-normalized and log-transformed. Principal component analysis (PCA) was performed using 50 principal components (PCs), with the top two PCs visualized using scanpy. Silhouette scores were computed using the scib_metrics package.

For evaluating organism classification accuracy by SRAgent, we used SRA BigQuery as ground-truth. Notably, manual assessment of misclassified datasets revealed that many were xenografts, so mis-classification was somewhat due to the constraint in SRA that "organism" must be a single value.

Statistical tests and metrics

Across the analyses presented in this study, several statistical tests and metrics were used, with details reported in the corresponding STAR Methods subsections and figure legends.

For the embedding evaluation experiments (Figure 4C), classification accuracy was computed on held-out test sets using 80%/10%/10% train/val/test splits, with MLP probes (two linear layers with one ReLU activation) trained for five epochs.

For the differential gene expression analysis (Figure 4D), ground truth differentially expressed genes (DEGs) were identified using the Wilcoxon rank sum test with a significance threshold of FDR < 0.05. Model performance was evaluated using AUROC and AUPR metrics, with ROC and PR curves generated by varying the classification threshold across effect score values.

For cell type validation, marker gene recovery was assessed by computing the percentage of PanglaoDB marker genes found among the top 200 upregulated genes per cell type category, using t-test scores from the scanpy rank_genes_groups function. Cell type classification accuracy was evaluated using row-normalized confusion matrices comparing predicted and ground-truth labels across 2,732,573 CELLxGENE cells (N = 2,652,156 cells for predictable categories; N = 80,417 cells for categories not in the training data).

For analytical factor evaluation, silhouette scores were computed on random subsets of N = 250,000 cells using the `scib_metrics` package. Cells with fewer than 100 unique genes and genes observed in fewer than 40 cells were excluded.

ADDITIONAL RESOURCES

Example analysis notebooks, including use cases for assembling a Huntington's disease atlas and querying embryonic data across species: https://github.com/ArcInstitute/scBaseCount_analysis.

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Supplemental information

scBaseCount: An AI agent-curated, standardized, auto-updated single-cell data repository

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1. Supplementary materials

1.1. Supplemental Figures

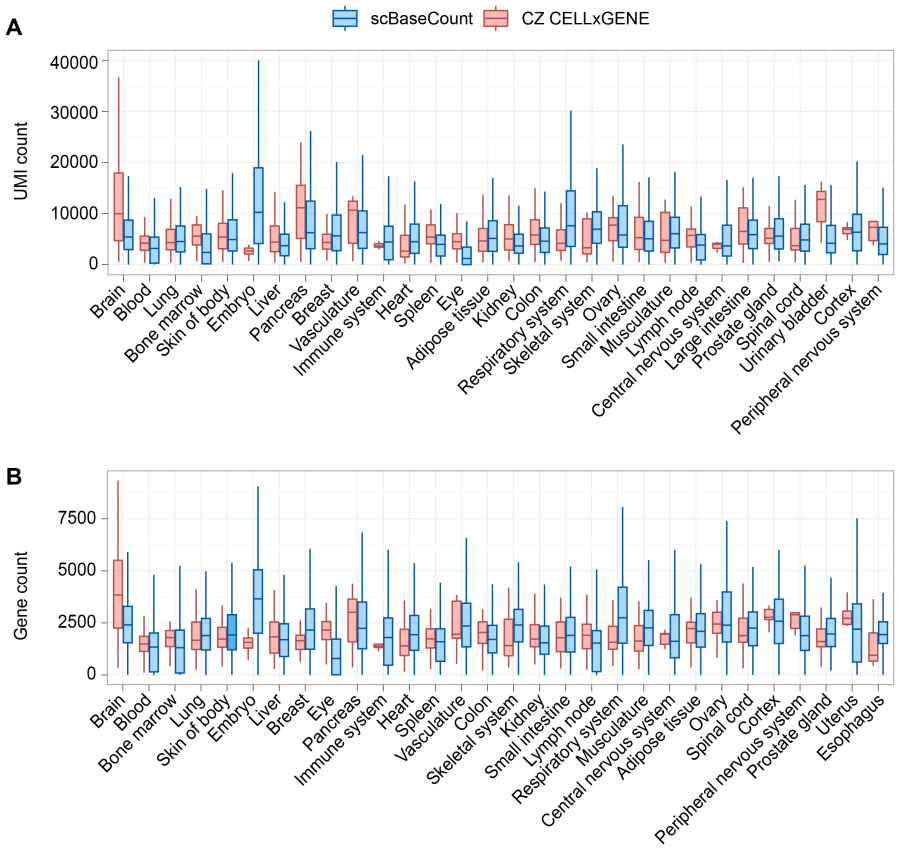


Figure S1: Count distribution comparison of scBaseCount to other publicly accessible single-cell data repositories, related to Figure 1. (**A-B**) Boxplots depicting the distribution of gene and UMI counts per cell across the top 30 tissues for scBaseCount (blue) and CZ CELLxGENE (red). The whiskers denote $1.5 \times IQR$. Just human and mouse cells were included.

Sample GSM3687217	Query DataSets for GSM3687217
Status	Public on Mar 27, 2019
Title	9 months APP/PS1 whole brain
Sample type	SRA
Source name	brain
Organism	Mus musculus
Characteristics	genotype/variation: APP/PS1 tissue: whole brain strain: APPPS1 B6-Tg(Thy1-APPswe; Thy1-PS1 L166P) platform: 10x Genomics number of cells: 3637
Extracted molecule	total RNA
Extraction protocol	All Methods listed in Van Hove et al. A single-cell atlas of non-parenchymal brain macrophages reveals unique transcriptional identities that are shaped by ontogeny and tissue environment. Nature Neuroscience. 2019.
Library strategy	RNA-Seq
Library source	transcriptomic
Library selection	cDNA
Instrument model	Illumina HiSeq 4000
Description	9 months APP/PS1 whole brain of APPPS1 B6-Tg(Thy1-APPswe; Thy1-PS1 L166P) mice
Data processing	Raw data bcl files were first demultiplexed using Cell Ranger mkfastq Demultiplexed data was then processed using the Cell Ranger count pipeline 8bp I7 sample barcode 26bp read1 (16bp Chromium barcode and 10bp UMI) 98bp read2 (transcript) Genome_build: mm10 Supplementary_files_format_and_content: tab-delimited text files including raw counts (output CellRanger Count) for each sample

Figure S2: An example of an experiment where the disease label differs between scBaseCount and CELLxGENE, related to Figure 2. In scBaseCount, this sample is labeled as "Alzheimer's", due to the APP/PS1 mouse model, which is commonly used to study Alzheimer's. In CELLxGENE, the cells from this experiment are labeled as healthy.

SRX22540254: GSM7902923: CV6, ADT; Homo sapiens; OTHER
1 ILLUMINA (Illumina NovaSeq 6000) run: 154.4M spots, 18.7G bases, 5.4Gb downloads

External Id: GSM7902923_r1

Submitted by: Koralov, Pathology, NYU Langone

Study: mRNA COVID-19 vaccine elicits potent adaptive immune response without the acute inflammation of SARS-CoV-2 infection [CV6]

PRJNA1040893 • SRP472232 • All experiments • All runs

hide Abstract

SARS-CoV-2 infection and vaccination elicit potent immune responses. Our study presents a comprehensive multimodal single-cell dataset of peripheral blood of patients with acute COVID-19 an of healthy volunteers before and after receiving the SARS-CoV-2 mRNA vaccine and booster. We compared host immune responses to the virus and vaccine using transcriptional profiling, coupled wit B/T cell receptor repertoire reconstruction. COVID-19 patients displayed an enhanced interferon signature and cytotoxic gene upregulation, absent in vaccine recipients. These findings were validate an independent dataset. Analysis of B and T cell repertoires revealed that, while the majority of clonal lymphocytes in COVID-19 patients were effector cells, clonal expansion was more evident among circulating memory cells in vaccine recipients. Furthermore, while clonal aß T cell responses were observed in both COVID-19 patients and vaccine recipients, dramatic expansion of clonal gdT cells w found only in infected individuals. Our dataset enables comparative analyses of immune responses to infection versus vaccination, including clonal B and T cell responses. Taken together, our comparative analysis shows that vaccination induces a robust adaptive immune response, including a durable clon B and T cell response, without the severe inflammation associated with infection. Overall design: We profiled crculating immune cells from five adults with acute COVID-19 infection and nine healthy adult seven of whom received theBNT162b2 vaccine. For three of the vaccine recipients, samples were als collected before and after receiving a booster. Samples were taken at multiple time points, resulting in total of 42 post-vaccination and 9 post-infection samples

Sample: CV6, ADT

SAMN38270960 • SRS19549909 • All experiments • All runs

Organism: Homo sapiens

Library:

Name: GSM7902923
Instrument: Illumina NovaSeq 6000
Strategy: OTHER
Source: OTHER
Selection: other
Layout: PAIRED
Construction protocol: PBMCs were isolated form whole blood using ficoll gradient. Cells were encapsulate in Gel-Beads-in emulsion (GEMs) using 10x Controller with a Chromium Next GEM Single Cell 5' Library & Gel Bead Kit v1.1 Gene expression library and VDJ libraries were constructed according to the manufacturer's protocol (10x Genomics) The CITE-seq and hashing protocol described by Stoeckius et. al was used to generate an antibody-derived tag (ADT) sequencing library and hashtag oligo (HTO) library (Stoeckius et. al., 2017). Gene expression library and VDJ libraries were constructed according to the manufacturer's protocol (10x Genomics)

Runs: 1 run, 154.4M spots, 18.7G bases, 5.4Gb

Run	# of Spots	# of Bases	Size	Published
SRR26844728	154,365,004	18.7G	5.4Gb	2023-11-21

ID: 30534529

Figure S3: An example of an experiment where the disease label differs between scBaseCount and CELLxGENE, related to Figure 2. In scBaseCount, this sample is labeled as 'Acute COVID', whereas in CELLxGENE, the cells from this experiment are labeled as 'normal'. This is assumed to be due to the fact that the control and treatment samples are not labeled in the metadata accessible to SRAgent.

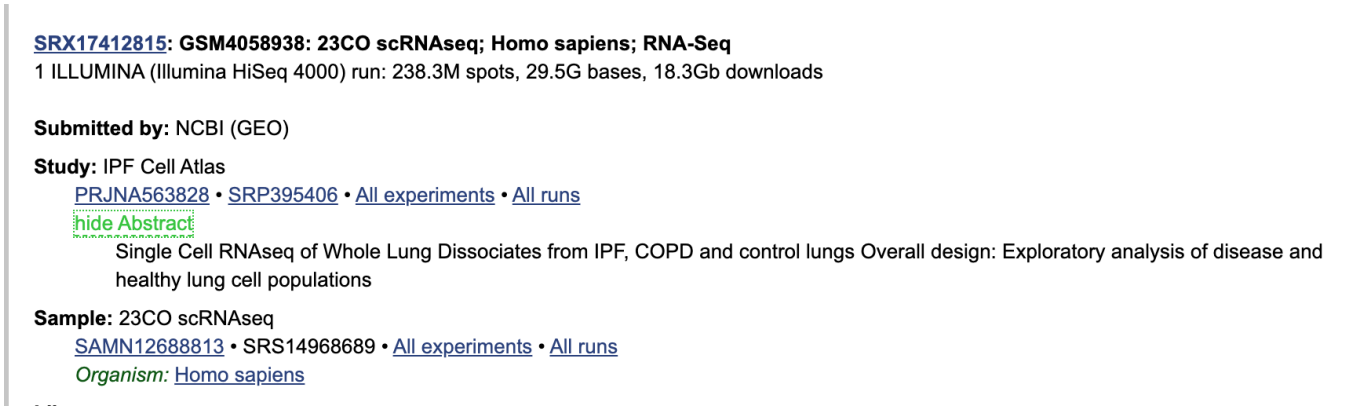


Figure S4: An example of an experiment where the disease label differs between scBaseCount and CELLxGENE, related to Figure 2. In scBaseCount, this sample is labeled as 'COPD', whereas in CELLxGENE, the cells from this experiment are labeled as 'normal'. This is assumed to be due to the fact that the control and treatment samples are not labeled in the metadata accessible to SRAgent.

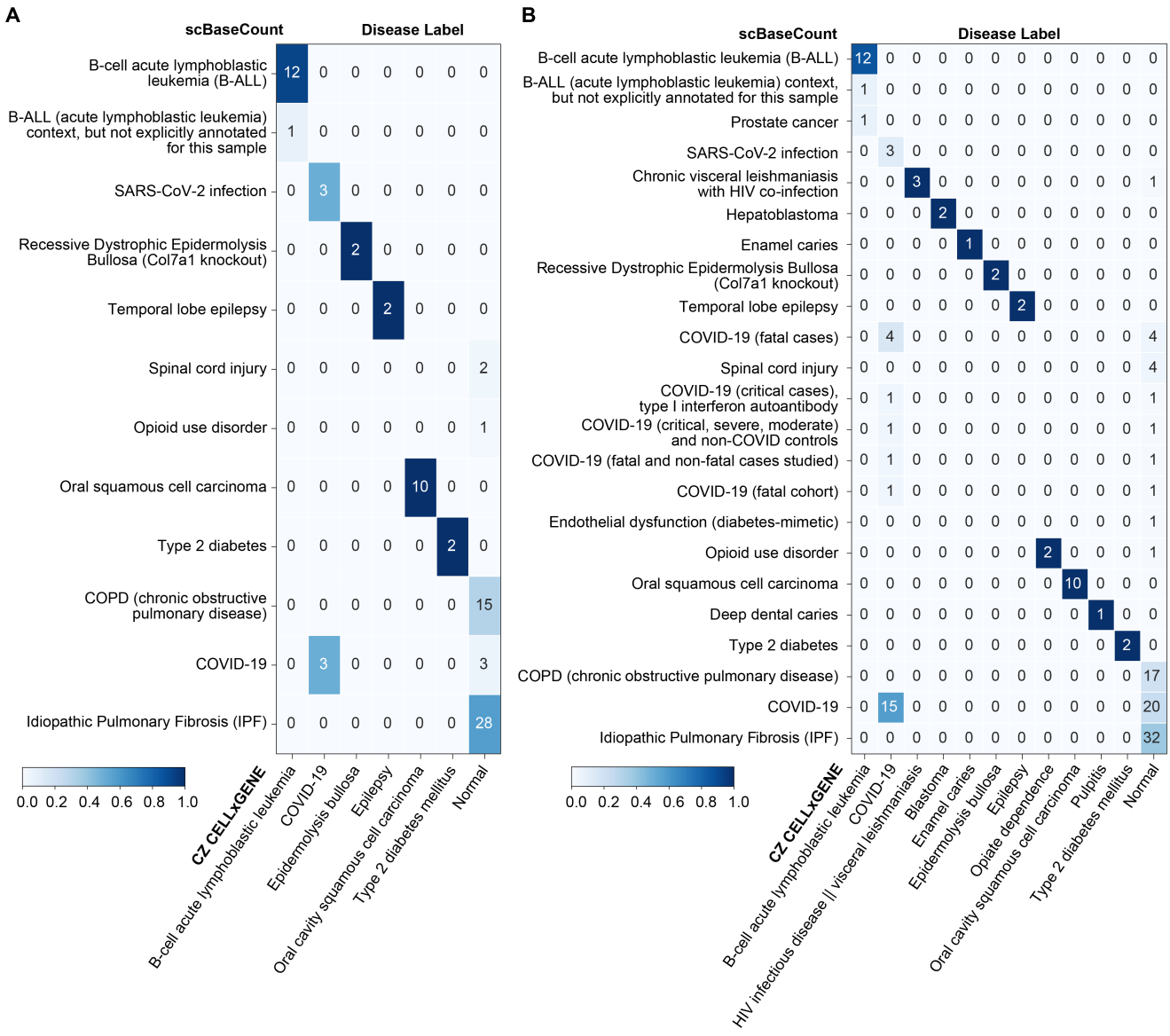


Figure S5: Confusion matrices of disease category labels between scBasecount and CZ CELLxGENE, related to Figure 2. (A) Matching of disease categories between scBaseCount and CZ CELLxGENE subsetted to high confidence annotations **(B)** Matching of disease categories between scBaseCount and CZ CELLxGENE across all annotations.

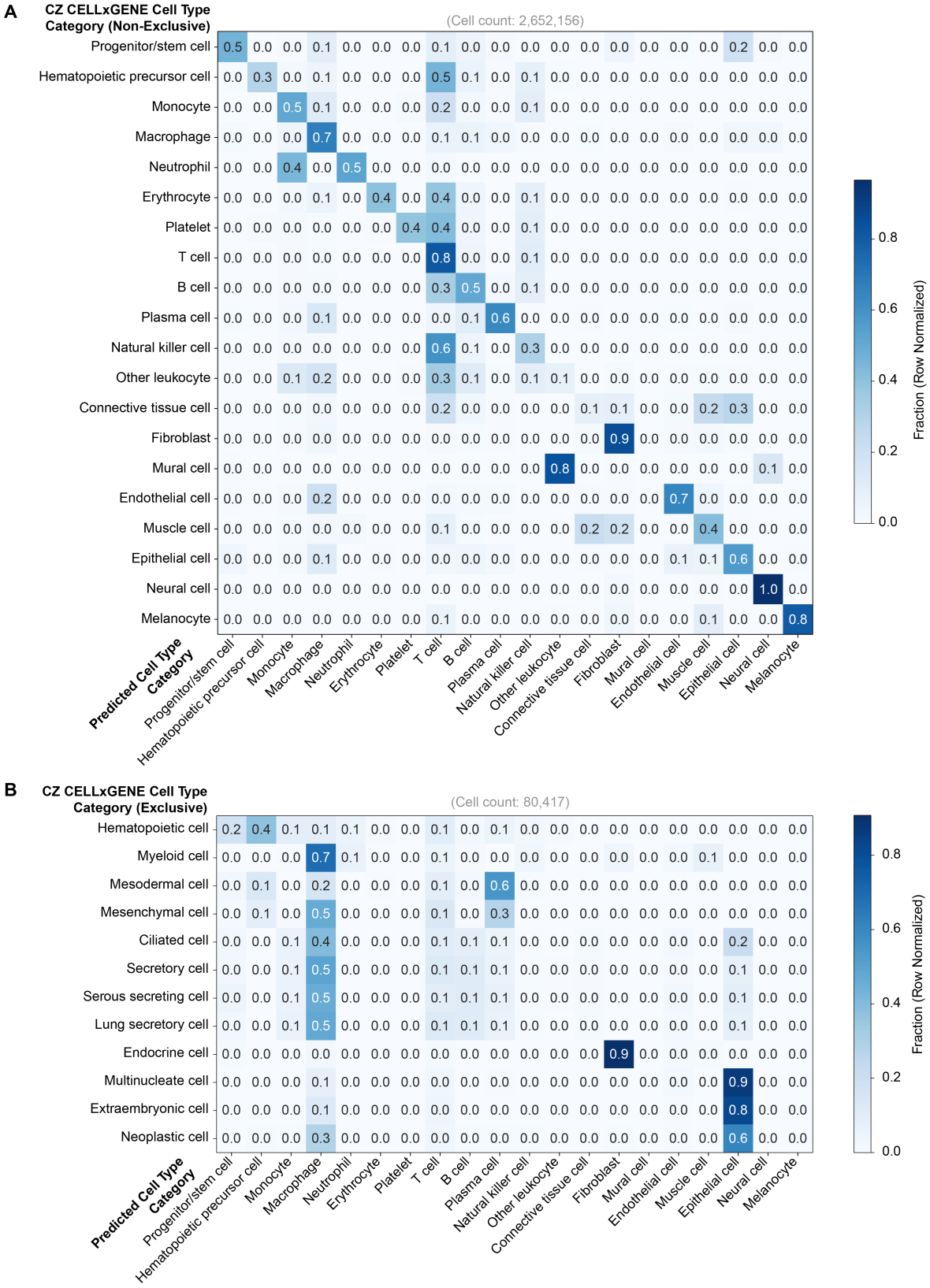


Figure S6: Confusion matrix of CELLxGENE cell type annotations versus scBaseCount predicted cell types, related to Figure 3. Both ground-truth (CELLxGENE) and predicted labels were mapped to parent cell types in the OBO Foundry Cell Ontology for coarser comparison (e.g., CD4-positive helper T cell to T cell), reducing 170 unique CELLxGENE labels and 77 predicted labels to shared parent categories. *(caption continued on next page)*

Figure S6: (*continued*) **(A)** Cell type categories present in both the Tabula Sapiens training data and the CELLxGENE subset (2,652,156 cells), demonstrating broad agreement between predicted and ground-truth annotations within categories that the classifier was trained on. **(B)** Cell type categories present only in CELLxGENE but absent from the Tabula Sapiens reference (80,417 cells), for which the classifier was forced to assign the closest available category; this panel highlights expected misassignments for cell types that lack a uniform transcriptomic signature in the reference (e.g., neoplastic cell, extraembryonic cell, multinucleate cell).

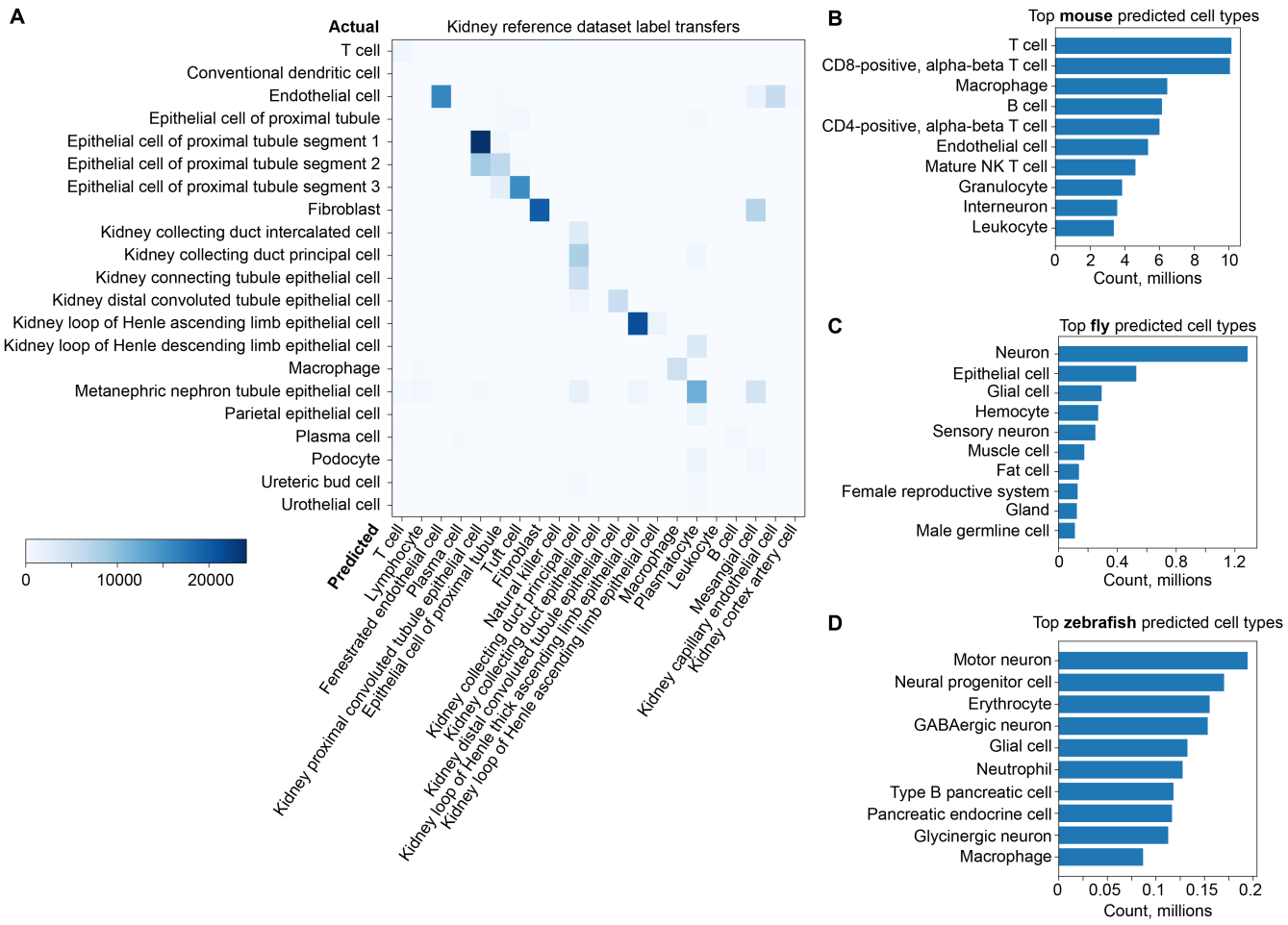


Figure S7: Cross-species cell type annotation, related to Figure 3. (A) Confusion matrix of automated mouse labels vs. human annotations for one CZI collection **(B)** Most common mouse cell types from our annotation pipeline **(C)** Most common fly cell types from our annotation pipeline **(D)** Most common zebrafish cell types from our annotation pipeline

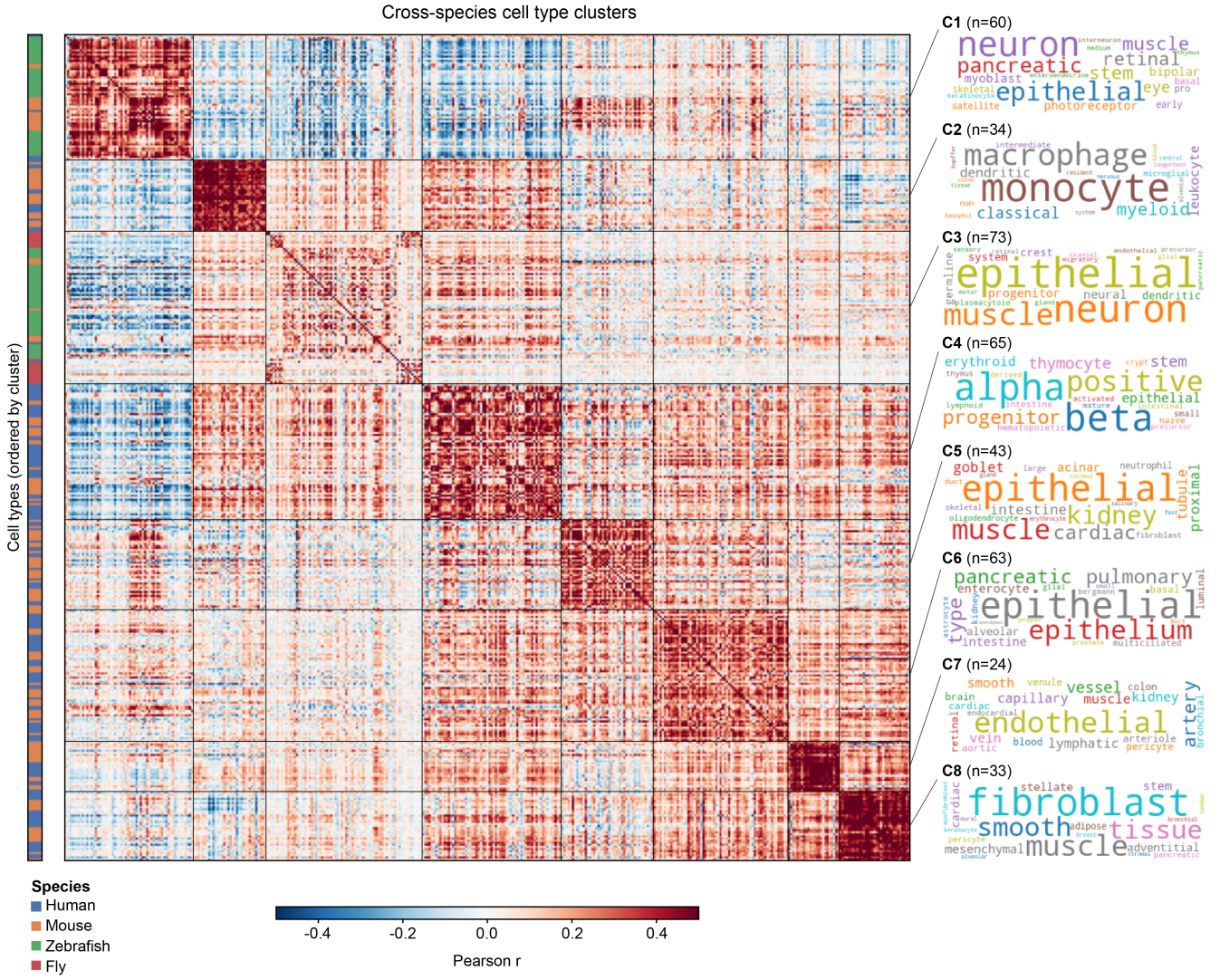


Figure S8: Cross-species cell type clusters, related to Figure 3. Clustering of cell types across species using Homo Sapiens homologous genes reveals species specific cell types, and shared cell type clusters across species. Progenitor cell types and neurons are generally shared among species, whereas immune cell types are more species-specific

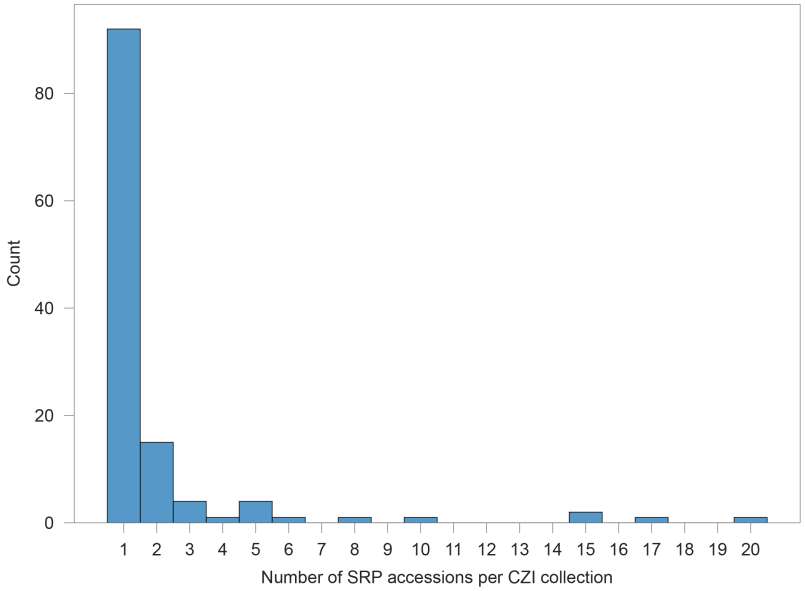


Figure S9: Histogram of the number of SRA studies (SRP) matching CZ CELLxGENE collection IDs, related to Figure 4. Since in most cases, there was a single SRP associated with a given CZI collection, we opted to treat these two variables as comparable in our data visualization in Figure 4.

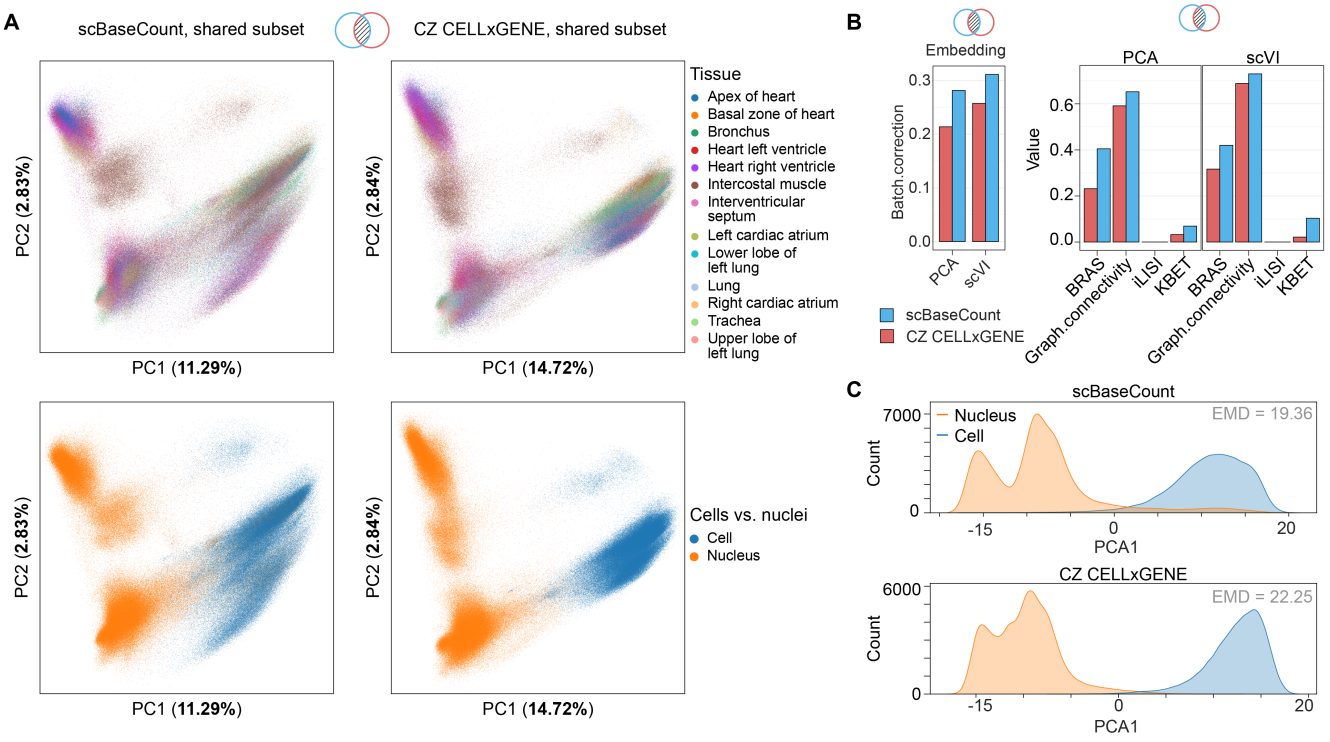


Figure S10: Paired comparison of single-cell and single-nucleus integration in scBaseCount versus CELLxGENE on 566,224 shared cells, related to Figure 4. (**A**) Principal Component Analysis (PCA) of a shared subset of cells present in both scBaseCount and CZ CELLxGENE that contain both single cell and single nuclei data (N=566, 224). Left: PCA plots of the shared subset in scBaseCount, colored by tissue type (top) and suspension type (bottom). Right: PCA plots of the same subset in CZ CELLxGENE, showing a greater separation between single-cell and single-nucleus samples along the first two principal components. (**B**) SCIB batch correction metrics computed over PCA and scVI, for tissue type, and sample suspension type (i.e. cells vs nuclei), confirming that sample suspension type is a stronger driver of variation in CZ CELLxGENE than in scBaseCount, and tissue type is better preserved in scBaseCount. (**C**) Distribution of cells and nuclei projected onto the first PC for scBaseCount and CZ CELLxGENE. We observed a more pronounced separation between cells and nuclei in CZ CELLxGENE as measured by Earth Mover's Distance (EMD).

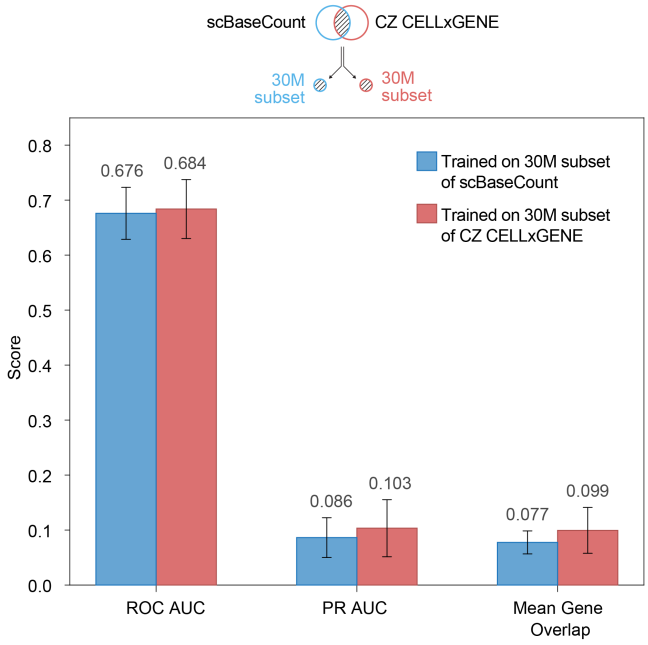


Figure S11: Performance of ablations of State Embedding models trained on matching subsets of CELLxGENE and scBaseCount data, related to Figure 4. The error bars denote the standard deviation. Values above each error bar are the mean.

1.2. Supplemental Tables

Dataset	Processing method	Cell prep	Publication
Integrated adult and foetal heart single-cell RNA sequencing	Demultiplexing, cell calling, alignment, and counts matrix generation were carried out using `Cell Ranger` version 6.1	Single cell, single nucleus (N=26, 972)	https://www.nature.com/articles/s44161-022-00183-w
A spatially resolved atlas of the human lung characterizes a gland-associated immune niche	Both types of libraries were mapped to an Ensembl 93-based reference (10x-provided GRCh38 reference, version 3.0.0). For nuclei samples, the reference was altered into a pre-mRNA reference as per 10x Genomics instructions	Single cell/single nucleus (N=142, 492)	https://doi.org/10.1038/s41588-022-01243-4
Cells of the adult human heart	Single-cell samples were mapped against the reference as it was provided. For single-nuclei samples, the reference for pre-mRNA was created using the 10x Genomics instructions.	Single cell/single nucleus (N=328, 595)	https://doi.org/10.1038/s41586-020-2797-4
Human skeletal muscle ageing atlas	Genomics skeletal muscle sequencing data were aligned and quantified using `Cell Ranger` (3.1.0) with GRCh38-3.0.0 human reference genome. Pre-mRNA version of reference genomes was used for alignment of nuclei datasets. `STARsolo` (STAR 2.7.3) '`-soloFeatures Gene GeneFull Velocyto`' was employed to separate spliced and unspliced counts, which were used to differentiate MF fragments.	Single cell, single nucleus (N=57, 581)	https://doi.org/10.1038/s43587-024-00613-3

Table S3: Overview of datasets, processing methods, and cell preparation techniques, related to STAR Methods. Shared human single cell data collections in scBaseCount and CZ CELLxGENE containing both single-cell and single-nucleus sequencing data. The number of cells/nuclei in each dataset are shown in parentheses in the "Cell prep" column. As tabulated here, for single cell nuclei, the authors had used pre-mRNA references for read mapping, and default `CellRanger mRNA` mapping for single cell RNA-seq datasets. This difference in the choice of analytical approach contributes to the separation of single-cell and single-nuclei studies.

Table S4: Comparison of SRA metadata extraction methods, related to STAR Methods. A random selection of 100 SRX accessions from the scBaseCount database were processed with each of the three methods. Values denote the number of accessions in which a metadata value was obtained.

Metadata Field	SRAgent	ffq	pysradb
Illumina platform?	100	100	83
Single-cell RNA-seq?	100	0	0
Paired-end sequencing?	100	0	100
Library preparation technology	100	0	0
10x Genomics chemistry version	100	0	0
Single-cell vs. single-nucleus	100	0	0
Organism	100	65	100
Tissue	100	0	0
Disease	100	0	0
Perturbation	100	0	0
Cell line	100	0	0
Tissue ontology ID (Uberon)	95	0	0